Supplementary Information

Assessment of computational methods for diabetic retinopathy lesion segmentation

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Supplementary Note 1: the registered model variants

One entry per registered variant: family, tier (1, main comparisons; 2, later rounds; 3, paper audit only), reproducibility level (R0 no code; R1 model definition only; R2 inference code and weights; R3 training code; R4 training code, weights, splits and environment), track (A, released weights; B, unified retraining of the official code; C, re-implementation from the paper), the code used, and what was changed or found when the code was run. The registry is `bench/configs/models.yaml` in the code repository; the notes are those recorded at the code audit and updated as runs completed.

General segmentation networks.

- **U-Net** (tier 1, reproducibility R3, track B); no author repo; use a reference PyTorch implementation.
- **DeepLabv3** (tier 1, reproducibility R3, track B); code: `official_code/DeepLabv3plus`; registered name is DeepLabv3. torchvision deeplabv3_resnet101 is ASPP plus bilinear upsampling, not the DeepLabv3+ decoder. The TensorFlow repo is the paper code and is not the network used here.
- **HRNet** (tier 1, reproducibility R3, track B); code: `official_code/HRNet-Semantic-Segmentation`.
- **nnU-Net** (tier 1, reproducibility R4, track B); code: `official_code/nnUNet`; train entry nnunetv2/run/run_training.py.
- **Swin-Unet** (tier 1, reproducibility R3, track B); code: `official_code/Swin-Unet`.
- **FCT** (tier 1, reproducibility R3, track B); code: `official_code/FCT`; official PyTorch port in PyTorch/main.py, not the TensorFlow entry. The wrapper widens the stem to 3 channels, assigns the four head channels to MA HE EX SE, and sets forward_size 384.
- **H2Former** (tier 1, reproducibility R3, track B); code: `official_code/H2Former`; general medical segmentation model used on several organs. Not a DR-specific method. The wrapper uses idrid_train.py for both IDRiD and DDR.
- **UNet++** (tier 2, reproducibility R3, track B); code: `official_code/Nested-UNet`.
- **UPerNet** (tier 2, reproducibility R3, track B); code: `official_code/UPerNet`.

Fundus-specific: multi-scale CNNs.

- **M2MRF** (tier 1, reproducibility R4, track AB); code: `official_code/M2MRF`; mmseg 0.8 / torch 1.6; weights on Google Drive; README: paper reports mean of 3 runs, code reports best.
- **HACDR-Net** (tier 1, reproducibility R3, track B); code: `official_code/HACDR-Net`; mmseg 0.30 / mmcv-full 1.6.2; no released weights found.

- **HRDecoder** (tier 1, reproducibility R4, track AB); code: `official_code/HRDecoder`; MICCAI 2024, mmseg 0.16 / mmcv 1.7, HRNet-W48 backbone; weights on Google Drive.
- **WSRFNet** (tier 1, reproducibility R1, track AB); code: `official_code/WSRFNet`; model.py + test.py, no train.py; weights not released; the training loop and the paper's loss are in bench/models/wsrfnet.py.
- **L-Seg** (tier 2, reproducibility R1, track C); code: `official_code/L-Seg`; Caffe prototxt and loss layer.
- **SAA** (tier 2, reproducibility R0, track C); code: `official_code/SAA`; no usable official code.
- **CARNet** (tier 3, reproducibility R0).
- **PMCNet** (tier 1, reproducibility R3, track B); code: `official_code/PMCNet`; train.py present; TF 1.13/Keras 2.3; no released weights; bundled data is one sample image; rewritten layer by layer in PyTorch (bench/models/pmcnet.py); the released dataGenerator one-hot-encodes 3 classes against the 5-class head.
- **MLNet** (tier 3, reproducibility R0).
- **LezioSeg** (tier 3, reproducibility R0).
- **HEDNet-cGAN** (tier 3, reproducibility R0).
- **Guo2024** (tier 3, reproducibility R0); Google Drive link of unknown content.
- **CLC-Net** (tier 1, reproducibility R0, track C); code: `reproductions/Wang2023_CLC-Net`; no author code; re-implemented from the paper (assumptions listed in the reproduction), trained through bench/models/clcnet.py.

Fundus-specific: relational and transformer models.

- **RTNet** (tier 2, reproducibility R0, track C); image + vessel input.
- **VTA** (tier 3, reproducibility R0).
- **TC-Net** (tier 1, reproducibility R0, track C); code: `reproductions/Zhang2023_TC-Net`; no author code; re-implemented from the paper (assumptions listed in the reproduction), trained through bench/models/tcnet.py.

Fundus-specific: semi-supervised.

- **SSMD-UNet** (tier 3, reproducibility R0); author split 70/5/25; EyePACS pretraining overlaps Retinal-Lesions source.
- **Zhou2019-CL** (tier 3, reproducibility R0).

Foundation models.

- **SAM** (tier 1, reproducibility R2, track B); code: `official_code/segment-anything`.
- **SAM2** (tier 1, reproducibility R3, track B); code: `official_code/sam2`.
- **SAM3** (tier 1, reproducibility R3, track B); code: `official_code/sam3`; train entry sam3/train/train.py; checkpoints are gated on Hugging Face.
- **MedSAM** (tier 1, reproducibility R3, track B); code: `official_code/MedSAM`.
- **SAMed** (tier 1, reproducibility R3, track B); code: `official_code/SAMed`.
- **SAM-Adapter** (tier 1, reproducibility R3, track B); code: `official_code/SAM-Adapter`.
- **SAM2-UNet** (tier 1, reproducibility R3, track B); code: `official_code/SAM2-UNet`.
- **RETFound** (tier 1, reproducibility R3, track B); code: `official_code/RETFound`; rmaphoh/RETFound_MAE, main_finetune.py is classification. A UPerNet head was added for segmentation. Weights on Hugging Face.
- **PerSAM** (tier 2, reproducibility R2, track B); code: `official_code/PerSAM`.
- **TP-DRSeg** (tier 3, reproducibility R0); code: `official_code/TP-DRSeg`; paper link 404; mirror has README only.
- **SAM2-DR-Chen2025** (tier 3, reproducibility R0).
- **Ophthalmology-SAM** (tier 3, reproducibility R0).

Supplementary Note 2: additional results

This note gives the detailed results that the main text summarises. Figure, table and Extended Data numbers refer to the main text.

Metric implementation. After the probabilities of the released M2MRF checkpoints are stored as 16-bit maps, the authors' function returns 0.647 on IDRiD and 0.426 on DDR, between the value on the float output and the exact AUPR. The paper reports the mean of three runs (0.672 and 0.489). Applied to the probability maps of the retrained models,, the 11-threshold implementation changed mAUPR on IDRiD by between −0.010 and +0.023 for the converged models, but on DDR, where the low-probability tail of the maps carries more weight, it added between 0.004 and 0.084 to them and 0.168 to the one model that had not converged: the inflation is model-dependent and is largest for the models whose maps are least calibrated, so it reorders models that differ by less than it (Fig. 2d,e; per-model values in Supplementary Table 5).

The literature audit also recorded inconsistencies between copied tables and their sources, such as transposed lesion columns and a value carried from one metric into another; they are listed per paper in Supplementary Data 2. Of the method papers, 2 use the exact average precision, 1 averages over images, and 5 say whether pixels are pooled over the test set or scores are averaged over images. Only 5 papers state that every baseline was rerun, and 10 say nothing about where their comparison numbers come from. DeepLabv3+ is reported on IDRiD by 11 papers with mAUPR from 0.48 to 0.67; copied values agree to the last digit, and the spread comes from the papers that reran a method under their own settings.

Author weights and author recipes. Rerunning the authors' pipeline is the direct test of whether released code reproduces a reported number. With the authors' batch size of 4 and their 40,000 iterations, three seeds of the M2MRF rerun on IDRiD reached 0.605–0.627 with the authors' metric (mean 0.615, s.d. 0.011) and 0.540–0.566 under our protocol (mean 0.552), against 0.676 and 0.639 for the released checkpoint and 0.672 for the three-run mean printed in the paper. The models produced by the released training pipeline were therefore 5–7 points below the released weights on every seed, a difference about five times the seed spread of the rerun itself and of the same architecture under the unified protocol (s.d. 0.010); a first rerun with an effective batch of 1 in place of 4 is reported separately in Fig. 3c and Table 3.

The released HRDecoder checkpoints reached 0.707 on IDRiD and 0.461 on DDR under our protocol; they were trained by their authors on a canvas twice as large as the unified one and on three to five times as many pixels. Rerun once with the authors' recipe (their geometry, augmentation, schedule and all 54 training images; one process in place of four), HRDecoder reached 0.672 with the authors' metric and 0.674 under our protocol on IDRiD, 0.04 and 0.03 below the released checkpoint (0.713 and 0.707); the checkpoint selection of the authors' pipeline, which evaluates on the test split, would have added 0.002 to the rerun and does not account for the difference. Applied to the independent test sets as released, the four checkpoints behaved like the retrained models (Fig. 3d): the DDR-trained checkpoints scored higher on the other pixel-labelled standard-field sources (IDRiD, the standard-field TJDR set) than on their own test set, kept 74–83% of their internal score on MAPLES-DR, and kept 36–60% of it on the coarsely or regionally labelled sets and on the ultra-widefield images, whereas the IDRiD-trained checkpoints fell on every other source (to 0.06–0.48 from 0.64 and 0.71).

Internal tests. On the internal tests (Fig. 4a and Table 4), adapted foundation models and a general-purpose high-resolution network led both sources, and the differences among them were within the seed-to-seed variation. HRNet-W48 reached 0.700 ± 0.004 mAUPR on IDRiD and 0.447 ± 0.009 on DDR (mean ± s.d. over three seeds), SAMed, a low-rank adaptation of SAM ViT-B trained on 512-pixel windows of the 1,440-pixel canvas, 0.706 ± 0.015 and 0.460 ± 0.027, and RETFound with a UPerNet head 0.699 ± 0.009 and 0.450 ± 0.018; the seed-0 margins of the two foundation models over HRNet-W48 on DDR (0.026 and 0.024) shrank to 0.013 and 0.004 over three seeds, within the seed scatter of the models themselves, and on IDRiD the three-seed margins (0.006 and −0.001) are within the width of the 27-image test set. SAM-Adapter, run with one seed, was level with SAMed on IDRiD (0.706) and 0.03 below it on DDR (0.433), and Swin-Unet (0.688 ± 0.003 and 0.431 ± 0.001) and the self-configuring nnU-Net (0.665 ± 0.008 and 0.431 ± 0.003) followed within 0.04 on both sources. The seven fundus-specific networks were below them: CLC-Net, the highest of the family on DDR (0.661 and 0.414), was 0.02 above U-Net on IDRiD and 0.04 above it on DDR, M2MRF reached 0.657 ± 0.010 and 0.369 ± 0.011, within 0.02 of its released checkpoint on IDRiD (0.639) and below it on DDR (0.401), and the other five spanned 0.54–0.65 on IDRiD and 0.23–0.36 on DDR (Table 4). The difference between the first two models of each column, less than 0.001 on IDRiD and 0.010 on DDR, is at most of the order of the median seed standard deviation of the table (0.010 and 0.009), whereas the span of the table (0.41 on IDRiD, 0.28 on DDR) is thirty to forty times larger. The 95% bootstrap intervals over test images (seed 0; Supplementary Table 4) are wider still: their median width across models is 0.08–0.12 AUPR for microaneurysms, hemorrhages and hard exudates on IDRiD and 0.07–0.19 on DDR, and 0.33 and 0.24 for soft exudates, which appear in few test images; the interval of the four-class mean, from the same resamples, has a median width of 0.10 on IDRiD and 0.09 on DDR, so the intervals of the leading models overlap (SAMed 0.42–0.51, RETFound with UPerNet 0.41–0.52, HRNet-W48 0.39–0.49 on DDR). IDRiD has 27 test images, and its intervals are correspondingly wide; we use it to connect to the literature and base comparisons between architectures on DDR and on the

independent tests.

Lesion classes differed more than models. The median AUPR over the 19 models was 0.87 for hard exudates and 0.44 for microaneurysms on IDRiD, and 0.60 and 0.15 on DDR; on DDR the highest microaneurysm score was 0.189 (0.14–0.24) for RETFound with UPerNet (seed 0; 95% bootstrap interval in brackets), against 0.155 (0.12–0.20) for HRNet-W48 and 0.154 (0.11–0.17) for nnU-Net, and the highest soft-exudate score 0.550 (0.42–0.65) for HRNet-W48; the intervals of the leading models overlap on every class.

Case study: one backbone, three decoders. HRDecoder, M2MRF and HRNet-W48 share the HRNet-W48 backbone, so their three rows compare a general head with two lesion-specific decoders on one backbone. Under the unified protocol the general head led on both sources (0.700 and 0.447), M2MRF followed (0.657 and 0.369) and HRDecoder was third (0.647 and 0.296), a span of 0.05 on IDRiD and 0.15 on DDR, whereas the released HRDecoder checkpoints score 0.707 and 0.461 under the same evaluation (see above). Three further runs separate the explanations. The authors' recipe, rerun once, gave 0.674 on IDRiD under our protocol, 0.03 above the unified run and 0.03 below the released checkpoint. Doubling the training amount of the unified protocol at the 1,440-pixel canvas did not close the gap (0.615 on IDRiD against 0.647 for the single budget, and 0.304 on DDR against 0.296, one seed each), whereas the same doubling raised the shared-backbone control HRNet-W48 on DDR from 0.447 to 0.469. The unified protocol at the authors' 2,880-pixel canvas and 1,920-pixel window, with the training amount counted in equivalent epochs at that canvas, gave 0.642, within 0.01 of the 1,440-pixel run (0.647); hard exudates and microaneurysms gained (0.853 against 0.823, 0.444 against 0.410) and hemorrhages and soft exudates lost (0.596 against 0.629, 0.676 against 0.728). Neither the canvas nor the training amount therefore accounts for the released checkpoint; what the authors' recipe adds beyond them, 0.03–0.06 on one seed, lies in the parts of the recipe that the protocol replaces (rotations of $\pm 45^{\circ}$ without photometric distortion, training on all 54 images without a validation split, and the last-step checkpoint; Supplementary Table 3).

Foundation models. Zero-shot use did not segment DR lesions. The automatic masks of SAM and SAM 2, with the density and area cap chosen on the validation pool, reached an any-lesion AUPR of 0.07 and 0.08 on IDRiD and 0.03 on DDR, against 0.78 for nnU-Net's maps on the same metric, and even when each mask was given the class it overlaps most in the ground truth the mAUPR stayed at 0.03. Text prompts to SAM 3 found two of the four classes: with the wording chosen on the validation pool, hard exudates reached an AUPR of 0.41 on IDRiD and hemorrhages 0.29 (0.24 and 0.23 on DDR), soft exudates 0.21 on IDRiD and 0.02 on DDR, and microaneurysms 0.01 under every wording. The wording mattered as much as the class, from 0.03 to 0.41 for hard exudates and from 0.00 to 0.21 for soft exudates, where the clinical term ("cotton wool spot") scored lowest and a descriptive wording ("white fluffy patch") highest (Fig. 5c). Prompts derived from the ground truth tell where the gap lies. With a box around every ground-truth component, SAM 2 reached 0.85 mAUPR on IDRiD, but an IDRiD image carries on average 44 microaneurysm and 142 hard-exudate components, and with 1, 5 and 20 boxes per image and class the mAUPR was 0.16, 0.35 and 0.55 (Fig. 5b); one positive point per component gave 0.12 on IDRiD and 0.02 on DDR. Given every box, the supervised models' own maps cut to the union of the same boxes scored 0.86 (nnU-Net) and 0.81 (M2MRF) on IDRiD and 0.71 and 0.67 on DDR, against 0.86 and 0.77 for SAM ViT-B, 0.85 and 0.76 for SAM 2 and 0.66 and 0.65 for MedSAM: inside a box the general promptable models delineated the lesion as well as the best supervised model on IDRiD and better on DDR. The gap of the promptable models therefore lies in finding lesions, not in drawing them.

The pretraining of the encoder was isolated under one UPerNet head, one recipe and one training amount, in two pairs fixed in advance: an encoder pretrained on retinal images against its own starting point, under each of two pretraining objectives (Fig. 5d,e; Supplementary Table 10). On DDR, RETFound's masked-autoencoder encoder reached 0.47 internally and 0.59 on the standard-field TJDR set, below the ImageNet masked autoencoder it was initialised from (0.52 and 0.63), and RETFound's DINOv2 encoder 0.49 and 0.62 against 0.51 and 0.62 for DINOv2 itself; the two pairs differed by -0.053 and -0.020 internally and the second pair by less than 0.015 on each of the three fine-grained independent sets. On IDRiD the same two pairs gave 0.71 against 0.73 and 0.71 against 0.74 (-0.021 and -0.028). Under one head, one recipe and one seed per encoder, retinal pretraining therefore added nothing to either starting point under either objective. The two natural-image encoders, ViT-L models that have never seen a fundus photograph, exceeded every row of Table 4 on both sources (0.52 and 0.51 on DDR against 0.469 for the best ranked model on seed 0; 0.73 and 0.74 on IDRiD against 0.717), by a margin larger than the seed scatter of any ranked model. These encoder cells were run as controls with one seed and one learning rate each and are reported in Supplementary Table 10 rather than as ranked rows.

Independent tests. Moving to a source that was never used for training did not by itself cost performance; the annotation tier and the imaging field did (Fig. 4b,c and Table 6; the 19 models on all eight test sets, seed means as in Table 4). The models trained on DDR averaged 0.369 mAUPR on the DDR test set and scored higher on the two other standard-field sources with pixel-level labels, 0.550 on IDRiD and 0.490 on the standard-field TJDR set (every model higher; two-sided Wilcoxon signed-rank $P < 0.001$), and the same on MAPLES-DR (0.372; 6 of 19 lower, $P = 0.595$). They lost a third to a half of their internal score wherever

the labels or the geometry differ from the training source: 0.211 on FGADR (−43%) and 0.193 on Retinal-Lesions (−48%), whose lesions are drawn as coarse regions, 0.251 pixel-level mAUPR against the region-level consensus of DiaRetDB1 (−32%), and 0.173 on the 133° ultra-widefield TJDR images (−53%; every model lower on each of the four sets, $P < 0.001$). The models trained on the 44 IDRiD images started higher internally (mean 0.630) and ended lower on every other source, fine-grained or not: 0.286 on DDR, 0.424 on the standard-field TJDR set and 0.350 on MAPLES-DR (−33% to −55%), and 0.172–0.239 on the coarse, region-level and ultra-widefield sets (−62% to −73%; every model lower on every set, $P < 0.001$). The 0.26 advantage of IDRiD over DDR on the internal tests therefore reversed outside the training source: on the six independent sets the DDR-trained checkpoint of a model was better than its IDRiD-trained one in 74 of 114 pairs, in 13–15 of 19 on each of the five standard-field sets; only on the ultra-widefield images were the IDRiD-trained checkpoints usually ahead (4 of 19 DDR-trained ahead), consistent with the larger field and higher resolution of their training images. For both training sources the pixel-labelled standard-field sets were the least affected (a gain from DDR, the smallest loss from IDRiD) and the coarse, region-labelled and ultra-widefield sets the most. On the ultra-widefield images the inference scale mattered less than the field itself: resizing them to the 1,440-pixel canvas instead of predicting at native resolution with sliding windows, the comparison variant of the registered plan, moved the median mAUPR by +0.02 from both sources (13 and 14 of 19 models higher at the smaller scale, differences from −0.12 to +0.13), far from the 0.2–0.4 that separate this set from the standard-field ones (Extended Data Fig. 3b).

Internal rank predicted independent rank where the labels were drawn the same way. For DDR-trained models the Spearman correlation between internal mAUPR and the score on IDRiD, the standard-field TJDR set and MAPLES-DR was 0.92, 0.88 and 0.82: HRNet-W48, third internally, was first on IDRiD (0.643) and on the standard-field TJDR set (0.616), SAMed, first internally, on MAPLES-DR (0.470) and FGADR (0.288; $\rho = 0.79$), and SAMed also led the mean of the three fine-grained independent sets (0.426; HRNet-W48 0.412). On Retinal-Lesions and DiaRetDB1 the correlation was 0.76 and 0.78 but the first place went to DeepLabv3 (0.282 and 0.348), tenth internally, whose output stride of 8 yields region-sized predictions, and on the ultra-widefield images the correlation vanished ($\rho = 0.01$) and the fully fine-tuned MedSAM, ninth internally, was first (0.243). For IDRiD-trained models the correlation was 0.75 on DDR and 0.71 on MAPLES-DR and 0.35–0.53 elsewhere; SAMed led DDR, the standard-field TJDR set and FGADR (0.368, 0.534, 0.280) and the fine-grained mean (0.394), whereas TC-Net, tenth internally, led MAPLES-DR (0.412), DeepLabv3, eighteenth, led Retinal-Lesions and DiaRetDB1 (0.260 and 0.311), and MedSAM, fourteenth, the ultra-widefield images (0.295). RETFound with UPerNet, second internally on DDR, scored lowest among the converged models on the ultra-widefield images from DDR (0.111): its 224-pixel windows see 1/19 of the 4,256-pixel canvas at a time.

Internal and independent performance were correlated across models (Spearman $\rho = 0.80$ for DDR-trained and 0.65 for IDRiD-trained models, internal mAUPR against the mean of the three fine-grained independent sets), and the model that led the fine-grained independent sets, SAMed (0.43 from DDR, 0.39 from IDRiD), was the same model as on the internal test. Averaged by family over both sources, the adapted foundation models were ahead on internal (0.55) and on fine-grained independent tests (0.38), the self-configuring baseline followed (0.54 and 0.36), and the fundus-specific (0.47 and 0.32) and general networks (0.47 and 0.30) were level with each other (Fig. 4d).

Training source and annotation granularity. The sources differ in how lesions were drawn before they differ in anything a model does (Fig. 6a; Methods). Counted as connected components of the prepared masks over every image of each source, an IDRiD image carries on average 43 microaneurysm and 144 hard-exudate components whose median diameter is 0.5% of the field-of-view width, and a DDR image 14 and 31 at 0.4%; MAPLES-DR, drawn pixel by pixel by retinologists, has 27 microaneurysms and 16 hard exudates per image at 0.8% and 1.0%. The two TJDR sets carry few components per image (3–7 per class) of the same small size (0.4–1.8%), so their difference to IDRiD and DDR is in how many lesions the images carry, not in how they were outlined. In the coarse and region-consensus sources the components are few and large: a Retinal-Lesions image carries 2 hard-exudate and 8 hemorrhage components with median diameters of 4.2% and 3.5% of the field, and a DiaRetDB1 image 1.5 and 4.5 at 4.1% and 2.9%, eight to nine times the hard-exudate diameter of IDRiD and DDR and two to four times their hemorrhage diameter, which for the same retina means that several lesions and the tissue between them were enclosed by one outline. FGADR, drawn as a mixture of single lesions and regions, sits between: 29 hard-exudate and 21 hemorrhage components per image, as many as DDR's, at 0.8% and 1.2%, somewhat larger than DDR's and a fifth to a third of Retinal-Lesions'.

To separate the effect of the training source from that of the model, the five representative models (U-Net, nnU-Net, M2MRF, SAM2-UNet and RETFound with UPerNet) were also trained on the standard-field TJDR images and on FGADR, at the pixel budget of DDR and with one seed, which together with the main table gives a matrix of four training sources by eight test sets (Fig. 6b,d; for the TJDR- and FGADR-trained models, whose sources fix no single field of view, the ultra-widefield set is predicted at the field-of-view scale, Methods). Every source served its own test set best: in-source training gave 0.670 mAUPR on IDRiD, 0.403 on DDR, 0.556 on the standard-field TJDR set and 0.414 on FGADR, against 0.396–0.584, 0.265–0.301, 0.450–0.523 and 0.179–0.232 from the other three sources (Fig. 6c). Training elsewhere cost 0.15 mAUPR on average over

the twelve off-diagonal cells and up to 0.27 (standard-field TJDR-trained models on IDRiD). Which foreign source was best depended on the labels of the test set rather than on the number of training images: DDR, the largest fine-grained source (383 images), was the best foreign source on every standard-field pixel-level test set (0.584 on IDRiD, 0.523 on the standard-field TJDR set, 0.391 on MAPLES-DR), whereas FGADR, three times larger (1,096 images) but drawn as mixed regions, was the best foreign source only on the two sets with coarse or region labels (0.224 on Retinal-Lesions and 0.263 on DiaRetDB1, against 0.201 and 0.259 from DDR) and on the ultra-widefield images (0.278 against 0.173 from DDR; its cells are predicted at the field-of-view scale, Methods), and lost 0.12 on average on the three fine-grained sources relative to in-source training; the standard-field TJDR set, the smallest source (172 images), transferred worst (0.396 on IDRiD, 0.265 on DDR).

We then added mixed or coarse labels to fine-label training while keeping the number of patch pixels fixed, for U-Net and M2MRF with one seed; differences smaller than twice the median seed standard deviation of the main table (0.018 on DDR) are reported without interpretation (Fig. 6e and Table 7). On the DDR test set, adding the 1,096 FGADR images to DDR changed mAUPR by −0.01 for U-Net (0.378 to 0.373) and −0.06 for M2MRF (0.380 to 0.320); adding only 383 FGADR images, as many as DDR has, by −0.01 and −0.04 (0.369 and 0.336), so the loss is not a matter of the fine images being outnumbered; adding Retinal-Lesions by −0.04 and −0.07 (0.341 and 0.306). On the IDRiD test set the same three training sets moved U-Net by +0.05, +0.03 and −0.03 (from 0.547) and M2MRF by −0.02, −0.02 and −0.04 (from 0.571), and on the standard-field TJDR and MAPLES-DR sets U-Net by −0.05 to +0.03 and M2MRF by −0.19 to −0.01. Mixed labels added to fine ones were therefore at best neutral and at worst costly, and coarse labels cost M2MRF 0.04–0.19 on every standard-field fine-grained test set. Trained on Retinal-Lesions alone, the two models scored 0.504 and 0.469 on its own test split but 0.164 and 0.104 on DDR and 0.248 and 0.167 on IDRiD; trained on FGADR alone, 0.364 and 0.437 on IDRiD.

A revision of the labels themselves, as opposed to a change of source, moved the scores far less. A refined annotation of the 27 IDRiD test images was released in 2026 (Refined IDRiD; one class per pixel, 1,024-pixel crops of the retinal region). We located each crop in the original image (every crop is the retinal bounding box padded to a square and resized; normalised cross-correlation with the original ≥ 0.9988 for all 27), resampled the official labels through the same geometry, and scored the test maps of the 19 IDRiD-trained models of Table 4 against both label sets without retraining or re-inference. The revision changed 13 of the 27 images for microaneurysms and hemorrhages, 14 for hard exudates and 6 of 14 for soft exudates, moving 5.7%, 1.6%, 4.0% and 0.7% of the labelled pixels (pooled intersection over union 0.94, 0.98, 0.96 and 0.99) and no component count except hard exudates (3,895 to 3,882). Under the same protocol the mAUPR of every model changed by between −0.002 and 0.000 (microaneurysm AUPR by −0.002 on average, the other classes by less than 0.002), and the order of the 19 models was unchanged (Kendall $\tau = 1.00$). The 1,024-pixel re-scoring itself stayed within 0.006 mAUPR of the main-table test at the 1,440-pixel canvas ($\tau = 0.95$). What separates the sources in Fig. 6 is therefore how lesions were defined and outlined, which a correction of individual masks within one style does not touch (Supplementary Table 9).

Lesion size, input resolution and training amount. Lesion size is the property that separates the four classes most clearly (Fig. 7a). On the training images the median diameter of a microaneurysm is 0.5% of the field-of-view diameter and that of a hemorrhage 1.1%; we froze these two values, computed on training images only, as the boundaries between small, medium and large lesions before the main-table training. Resizing a whole image to a small input removes lesions from the training target itself: at 224 pixels 16.0% of the microaneurysms and 25.8% of the hard exudates of the IDRiD training set no longer cover a single pixel, and on DDR 34.3% and 35.3%; at 384 pixels the DDR figures are 11.5% and 16.9%, and on the 1,440-pixel canvas of the unified protocol essentially none are lost (Fig. 7b).

Lesion-level recall followed size for every model (Fig. 7c). On the DDR test set the median recall over the 19 models (seed 0, threshold chosen on the validation set) was 0.26 for small, 0.49 for medium and 0.66 for large lesions; the best models detected 0.30 of the small ones (Swin-Unet and SAM2-UNet, nnU-Net ResEnc-M 0.29) and DeepLabv3, decoding at an output stride of 8, 0.07. By class, the median recall of small lesions was 0.46 for microaneurysms, 0.34 for hard exudates and 0.15 for hemorrhages, and the median lesion-level precision 0.29, 0.73 and 0.65. Input resolution acted mainly through this channel. Shrinking the canvas from 1,440 to 512 pixels at an equal pixel budget lowered mAUPR on DDR from 0.380 to 0.353 for M2MRF and from 0.466 to 0.441 for RETFound with UPerNet (one seed), the recall of small lesions from 0.27 to 0.17 and from 0.28 to 0.10, and the microaneurysm AUPR from 0.148 to 0.111 and from 0.189 to 0.083 (Fig. 7d). Patches at native resolution, where a 960-pixel patch covers a smaller part of the retina and the fixed pixel budget buys fewer passes over each image, raised the small-lesion recall of M2MRF to 0.33 and of RETFound to 0.29, but lowered mAUPR to 0.312 and 0.448: for RETFound the microaneurysm and hemorrhage AUPR rose (0.189 to 0.197 and 0.491 to 0.502) while the hard- and soft-exudate AUPR fell (0.640 to 0.620 and 0.546 to 0.471), so the two smallest classes gained from native resolution and the two largest lost.

More training images helped, and the two models gained at different points of the curve. With 10%, 50% and 100% of the DDR training set (38, 192 and 383 images; equal number of patch pixels, so each image is seen ten, two and one times as often), mAUPR on the DDR test set was 0.321, 0.373 and 0.378 for U-Net and 0.296, 0.315 and 0.380 for M2MRF, and on IDRiD 0.492,

0.540 and 0.547 and 0.445, 0.544 and 0.571 (Fig. 7e): U-Net had most of its gain with the first half of DDR, M2MRF with the second half. Merging the three fine-grained sources IDRiD, DDR and standard-field TJDR (599 images) at the same budget gave 0.367 and 0.368 on DDR, within 0.02 of DDR alone, and raised the scores on the sets outside the training sources by 0.00–0.04 (MAPLES-DR 0.372 and 0.349 against 0.329 and 0.324; FGADR 0.192 and 0.224 against 0.170 and 0.212; Retinal-Lesions 0.177 and 0.153 against 0.142 and 0.150), the gain of more fine-grained images rather than of in-source data. A source merged into the training set raised the score on its own test set by more, 0.05–0.07 (standard-field TJDR 0.562 and 0.565 against 0.494 and 0.506; IDRiD 0.607 and 0.619 against 0.547 and 0.571), which is the in-source gain of the training-source matrix and not generalization; these two tests are in-source for the merged run and are marked as such in Fig. 7e. Exchanging the loss of U-Net with the architecture fixed moved mAUPR on DDR between 0.357 (cross-entropy alone) and 0.378 (cross-entropy with batch Dice, the main-table recipe; seed 0, as for every loss cell); Dice alone gave 0.237, the per-image Dice of the registered recipe 0.119, the rare-class collapse described in the main text (Extended Data Fig. 2).

Disease stage of the test images. Scores also depend on the disease stage of the test images, because the pooled AUPR follows the number of lesion pixels available. On the four test sets that carry a DR grade per image, every score was recomputed within grades 0–1 and within grades 2–4 with the thresholds of the full test (Methods; Supplementary Table 12). On the DDR test set the median mAUPR of the DDR-trained models (seed 0) was 0.21 on the 20 images of grades 0–1 and 0.36 on the 203 images of grades 2–4, on FGADR 0.04 and 0.22 (94 and 459 images) and on Retinal-Lesions 0.07 and 0.17 (150 and 328); the image-level AUC on DDR was 0.54 within grades 0–1, where few images carry a lesion of the class (defined for 12 of the 19 models), and 0.75 within grades 2–4. MAPLES-DR is graded on the Canadian teleophthalmology scale, under which most of its images are R1 (mild) although they were drawn from MESSIDOR grades 1–3, so 56 of its 60 test images fall in grades 0–1 (median 0.33) and only 4 in grades 2–4 (0.40), too few for a comparison. The grade mix of a test set therefore sets the level of every model's score, and the comparisons of this study hold within a test set and not across sets.

Human reference. DiaRetDB1 gives the comparison a human reference. Each of its four experts, scored against the consensus of the other three with the same protocol, reached a pixel-level mAUPR of 0.15–0.28 (median 0.21) and an image-level AUC of 0.87–0.93 (Table 6, bottom block, and Supplementary Table 13; a four-level confidence map yields a four-point precision–recall curve, so the pixel-level value is a lower bound). The retrained models scored against the four-expert consensus (seed 0) spanned 0.09–0.32 pixel-level mAUPR (median 0.28 from DDR, 0.25 from IDRiD) and 0.71–0.94 image-level AUC (median 0.84 and 0.83), so on pixel-level agreement with the consensus all but 3 of the 38 model–source pairs lie within or above the band of the experts, whereas on image-level detection only 7 (DDR-trained) and 5 (IDRiD-trained) of 19 reach the experts' lowest value of 0.87; the experts' own microaneurysm agreement was the lowest of the four classes (AUPR 0.001–0.022 against 0.039 for the median model), which says as much about the consensus rule as about the models.

Ensembles and reference model. Averaging the probability maps of the retrained models, in compositions fixed before any ensemble was scored (Methods), gave the highest scores of the study on every test set from DDR and on all but one from IDRiD (Table 6, lower block; seed-0 members). From DDR, the mean of all 18 members reached 0.505 on the DDR test set and the mean of the five best by validation mAUPR 0.511, against 0.469 for the best member; the same two ensembles reached 0.677 and 0.680 on IDRiD (best member 0.645), 0.637 and 0.652 on the standard-field TJDR set (0.624) and 0.496 and 0.506 on MAPLES-DR (0.466). On the coarsely and region-labelled sets the gain was smaller (0.298 and 0.290 on FGADR and Retinal-Lesions for the five best members against 0.290 and 0.278 for the best member; 0.335 on DiaRetDB1 against 0.325). The gain over the best member was thus 0.02–0.06 mAUPR on the fine-grained test sets and -0.01 to 0.02 on the others, and the four adapted foundation models averaged by themselves matched or exceeded the ensemble of all models (0.509 on DDR, 0.676 on IDRiD, 0.657 on the standard-field TJDR set). On FGADR and Retinal-Lesions the two compositions that contain HACDR-Net were computed without it (its maps for those two sets were not available) and are marked in the table. Ensembles are a reference for users and do not enter the rank tests above.

A reference model that follows these findings, RetiSeg-Ref, averages the two main-table models with the highest DDR validation score (RETFound with UPerNet and SAMed), each trained once on the three fine-grained sources merged (599 images; Methods), and will be released with the benchmark. Trained on the merged sources, the two members reached 0.470 (SAMed) and 0.468 (RETFound with UPerNet) on the DDR test set and their mean 0.501, against 0.469 for the best single DDR-trained model (SAMed, seed 0) and 0.505 for the mean of all 18 DDR-trained members. On the five test sets outside its training sources the averaged model reached 0.482 on MAPLES-DR, 0.309 on FGADR, 0.278 on Retinal-Lesions, 0.339 on DiaRetDB1 and 0.302 on the ultra-widefield images at the field-of-view scale, within 0.02 of the 18-member DDR-trained ensemble on each of the four standard-scale sets (above it on FGADR, below on MAPLES-DR; 0.291, 0.278, 0.335 and 0.496) with two members in place of 18; on the test sets of the two other merged sources, which are in-source for it, it reached 0.710 on IDRiD and 0.696 on the standard-field TJDR set, above every DDR-trained model and ensemble there but below the IDRiD-trained ensemble

on IDRiD (0.764), whose members saw each IDRiD image about four times as often under the IDRiD budget. Its scores are a starting point for users of the benchmark and are not ranked (Supplementary Table 11).

Computational cost. Because the training amount is fixed in pixels, the cost of a model under the unified protocol reflects its throughput and its patch size: one seed on IDRiD took from 1.5 GPU hours (HRDecoder and U-Net) to 20 (FCT), and a DDR seed 3.9 times as long (median ratio), the slowest being the models that process small patches with sliding windows (FCT, 72 hours on DDR; SAM-Adapter 26, HACDR-Net 28 and CLC-Net 39 hours). nnU-Net follows its own schedule of 1,000 epochs rather than the pixel budget (7 hours per source, 11–13 for the residual-encoder plan). The two models that led the tables were not the costliest to train: SAMed took 2.1 and 7.4 hours and RETFound with UPerNet 6.1 and 24 hours per seed. In the inference audit (15 of the 19 models; Methods), one forward pass takes 443 GFLOPs (PMCNet) to 20,870 (RETFound with UPerNet, whose 224-pixel windows tile the canvas), 0.03–1.10 s per image on an A100 and 0.3–8.7 GB of GPU memory. Here the two leading models differ by an order of magnitude (SAMed 0.30 s, RETFound with UPerNet 1.10 s per image, against 0.05 s for HRNet-W48 and 0.03 s for U-Net), so the general-purpose high-resolution network gives the same accuracy at a twentieth of the inference time of the ViT-L model. Accuracy was not related to training cost, inference cost or parameter count (Extended Data Fig. 4 and Supplementary Table 8).

Supplementary Tables

Supplementary Table 1 Literature audit: the experimental setup stated by each method paper in the ledger. Split, input resolution, AUPR implementation and the provenance of baseline numbers as extracted from the paper's text and tables (literature/meta/paper_audit.tsv in the repository holds the verbatim evidence for every cell).

Paper	Year	Venue	Datasets	IDRiD split	DDR split	Input resolution	AUPR impl.	Pooling	Baselines	*n*
Guo2019 L-Seg	2019	Neurocomputing	IDRiD;e-ophtha;DDR	official	official	IDRiD and e-ophtha: 'resize each image and…	11-threshold	not stated	rerun all	4
Zhou2019 collaborative semi s…	2019	CVPR	IDRiD	official	NA	640x640 (Fig. 2 shows 640*640*3 input / 640…	not stated	not stated	not stated	5
Xiao2020 HEDNet cGAN	2020	arXiv (2007.138…	IDRiD	official	NA	original 4288 x 2848; training: 'we randoml…	exact	pooled ('…	rerun all	2
Bo2022 SAA	2022	ISBI	IDRiD;DDR	official	official	IDRiD: 'resized images and their correspond…	not stated	not stated	not stated	6
Guo2022 CARNet	2022	Complex & Intel…	IDRiD;e-ophtha;DDR	official	official	'all images are uniformly resized to 1024 x…	NA	not stated	rerun all	8
Hasan2022 DRG-Net	2022	arXiv (2212.146…	IDRiD;FGADR	official	NA	'For all experiments, the sizes of the inpu…	not stated	not stated	copied	8
He2022 PMCNet	2022	arXiv (2205.157…	DDR;IDRiD;e-ophtha	official	official	IDRiD and E-ophtha: 'resized each image and…	not stated	not stated	rerun some	11
Huang2022 RTNet	2022	IEEE Transactio…	IDRiD;DDR	official	official	training: large images random resized and c…	exact	not stated	rerun some	7
Ali2023 LezioSeg	2023	Electronics (MD…	IDRiD;e-ophtha;DDR	official	official	training: each image split into 4 non-overl…	per-image mean	per-image	not stated	11
Geetha2023 RILBP-YNet	2023	Biomedical Sign…	IDRiD;e-ophtha (plu…	official	NA	black border cropped by contour analysis, t…	11-threshold	not stated	copied	11
Liu2023 M2MRF	2023	Pattern Recogni…	DDR;IDRiD	official	official	DDR: downsample so long side = 1024, zero-p…	11-threshold	pooled	rerun some	12
Qiu2023 Learnable Ophthal-molo…	2023	arXiv	IDRiD	not stated	NA	not stated	NA	not stated	not stated	1
Ullah2023 SSMD-UNet	2023	Sci Rep	FGADR;IDRiD	official	NA	'bicubic interpolation to resize the images…	not stated	not stated	rerun some	9
Wang2023 CLC-Net	2023	Neurocomputing	IDRiD;e-ophtha;DDR	official	official	no global resize: Otsu to remove non-tissue…	not stated	not stated	rerun some	8
Wang2023 VTA hyperbolic	2023	Scientific Repo…	IDRiD;DDR	official	official	not stated (ViT patch size 16; model pre-tr…	not stated	not stated	rerun all	4
Yin2023 DB-UNet	2023	IEEE Access	IDRiD;DDR;e-ophtha…	official	official	IDRiD: ROI 2560x3840 downscaled by t (t=2 f…	11-threshold	not stated	rerun some	8
Zhang2023 TC-Net	2023	Comput Biol Med	IDRiD;DDR;ISIC-2017	not stated	own	all images resized to 512x512	NA	not stated	not stated	7
Bian2024 MLNet	2024	Algorithms (MDP…	DDR;DiaRetDB1;IDRiD	official	official	black borders cropped, images uniformly res…	not stated	not stated	rerun some	7
Fu2024 LFRC-Net	2024	Applied Science…	IDRiD;DDR;Local ult…	official	official	IDRiD: cropping, zero-padding and resizing…	not stated	not stated	rerun some	12
Guo2024 deep multiscale	2024	SSRN preprint (…	IDRiD;DDR;DiaRetDB1…	official	official	three input scales 384x384, 512x512, 1024x1…	not stated	pooled	not stated	5
Li2024 TP-DRSeg	2024	MICCAI	IDRiD;DDR	not stated	not stated	resized to 1024x1024 ('We resize the input…	not stated	not stated	not stated	8
Li2024 WSRFNet	2024	IJCAI	IDRiD;DDR	official	official	resized to 1440x960 (IDRiD) and 1024x1024 (…	11-threshold	pooled	rerun some	8
Playout2024 cross dataset gen…	2024	arXiv (ISBI-sty…	IDRiD;DDR;FGADR;Ret…	official	official	cropped to remove black background, resized…	11-threshold	not stated	copied	3
Xu2024 HACDR-Net	2024	AAAI	IDRiD;DDR	official	official	IDRiD resized to 1440x960 (following M2MRF)…	not stated	not stated	not stated	13
Xu2024 MSAG	2024	Scientific Repo…	IDRiD;DDR	official	not stated	IDRiD: 4288x2848 cut into 54 blocks of 512x…	not stated	not stated	not stated	9
Xu2024 PGAFT	2024	Sci Rep	IDRiD;DDR	official	official	'both the larger color fundus images and va…	not stated	not stated	rerun all	11
Chen2025 SAM2 DR	2025	ISBI	IDRiD;DDR	official	official	resized to 1024x1024 ('Input images are res…	not stated	not stated	not stated	6
Guo2026 TVFNet	2026	Neural Comput A…	IDRiD;DDR;FGADR;e-o…	official	official	'the fundus images were uniformly resized t…	not stated	not stated	rerun some	18
Qi2026 TEDR-YNet	2026	IEEE Access	IDRiD;DDR;TJDR	official	own	images from all datasets were resized to 51…	not stated	not stated	rerun some	10
Wang2026 CSIF	2026	Applied Soft Co…	DDR;IDRiD	official	official	DDR: long edge scaled to 1024 then zero-pad…	not stated	not stated	rerun some	11

"not stated" means the paper gives no information. Baselines: how the comparison numbers were obtained (rerun, copied, mixed, not stated). *n*, number of compared methods.

Supplementary Table 2 One method, many reported values. Every method that appears with a mean AUPR on IDRiD or DDR in at least two papers of the ledger: the number of papers, the lowest and highest value, their spread, and how many of the values were produced by the reporting paper (rerun), carried from an earlier paper (copied) or cite a source outside the ledger (unverifiable).

Method	Dataset	Papers	Min	Max	Spread	Rerun	Copied	Unverif.
AIMIA	IDRiD	3	.437	.437	.000	0	0	0
Attention U-Net	DDR	2	.198	.389	.191	2	0	0
Attention U-Net	IDRiD	2	.418	.549	.132	2	0	0
CASENet	DDR	3	.193	.193	.000	1	0	0
CASENet	IDRiD	4	.482	.482	.000	1	0	0
CE-Net	DDR	2	.309	.504	.195	2	0	0
CE-Net	IDRiD	2	.470	.635	.165	2	0	0
CIL	IDRiD	3	.535	.535	.000	0	0	0
DNL	DDR	2	.401	.401	.000	1	0	0
DNL	IDRiD	2	.591	.591	.000	1	0	0
Deep-multiscale (Guo2024)	DDR	3	.349	.353	.004	0	0	0
Deep-multiscale (Guo2024)	IDRiD	2	.710	.710	.000	0	0	0
DeepLabv3+	DDR	10	.290	.452	.162	6	0	0
DeepLabv3+	IDRiD	11	.483	.666	.183	6	0	0
DenseUNet	DDR	2	.336	.483	.147	1	0	0
DenseUNet	IDRiD	2	.564	.651	.087	1	0	0
EAD-Net	IDRiD	2	.549	.549	.000	0	0	0
FCRN	DDR	2	.096	.096	.000	1	0	0
FCRN	IDRiD	3	.455	.455	.000	1	0	0
FCT	DDR	2	.250	.250	.000	0	0	0
FCT	IDRiD	2	.431	.431	.000	0	0	0
HED	DDR	6	.191	.430	.239	2	0	0
HED	IDRiD	7	.366	.639	.274	2	0	0
HRNet	DDR	4	.452	.483	.031	3	0	0
HRNet	IDRiD	4	.649	.689	.040	3	0	0
HRNet+OCR	DDR	2	.452	.474	.022	0	0	0
HRNet+OCR	IDRiD	2	.649	.660	.011	0	0	0
L-Seg	DDR	13	.321	.321	.000	0	0	0
L-Seg	IDRiD	14	.399	.661	.262	0	0	0
M2MRF	DDR	9	.489	.499	.010	0	0	0
M2MRF	IDRiD	9	.583	.672	.089	0	0	0
MedLabs	IDRiD	3	.440	.440	.000	0	0	0
MedSAM	DDR	2	.193	.193	.000	0	0	0
MedSAM	IDRiD	2	.266	.266	.000	0	0	0
PMCNet	DDR	4	.364	.364	.000	0	0	0
PMCNet	IDRiD	3	.681	.681	.000	0	0	0
PSPNet	DDR	4	.177	.392	.215	3	0	0
PSPNet	IDRiD	4	.337	.587	.250	3	0	0
RILBP-YNet	IDRiD	3	.659	.659	.000	0	0	0
RTNet	DDR	4	.336	.336	.000	0	0	0
RTNet	IDRiD	3	.421	.708	.287	0	0	0
SAA	DDR	4	.366	.410	.044	0	0	0
SAA	IDRiD	5	.648	.674	.026	0	0	0
SAM	DDR	2	.003	.003	.000	0	0	0
SAM	IDRiD	2	.006	.006	.000	0	0	0
SAM-Adapter	DDR	2	.299	.299	.000	0	0	0
SAM-Adapter	IDRiD	2	.346	.346	.000	0	0	0
SAMed	DDR	2	.300	.300	.000	0	0	0
SAMed	IDRiD	2	.354	.354	.000	0	0	0
SDNU	IDRiD	3	.477	.477	.000	0	0	0
SOONER	IDRiD	3	.554	.554	.000	0	0	0
Swin (UPerNet)	DDR	5	.468	.486	.018	2	0	0
Swin (UPerNet)	IDRiD	4	.649	.690	.042	2	0	0
Twins-SVT-B	DDR	2	.461	.461	.000	1	0	0
Twins-SVT-B	IDRiD	2	.638	.638	.000	1	0	0
U-Net	DDR	7	.190	.429	.239	2	0	0
U-Net	IDRiD	6	.347	.652	.305	2	0	0
U-Net++	DDR	4	.192	.409	.217	3	0	0
U-Net++	IDRiD	3	.391	.572	.181	3	0	0
VRT	IDRiD	8	.647	.647	.000	0	0	0
iFLYTEK-MIG	IDRiD	7	.648	.648	.000	0	0	0

Values as printed, normalised to [0, 1]. The full extraction, one row per paper, baseline, dataset, class and metric, is literature/meta/reported_results.tsv.

Supplementary Table 3 Training amount, batch and selection of every main-table run (seed 0). Steps are the shared pixel budget expressed in the model's patch size (Methods); the loader batch is the largest that fits a 40 GB card and is accumulated to the recipe's effective batch; the selected checkpoint is the validation mAUPR maximum over 20 evaluations. nnU-Net follows its own 1,000-epoch schedule and planning.

Model	Source	Steps	Loader batch	Accum.	Eff. batch	Patch	LR	Best val. mAUPR	GPU h
U-Net	IDRiD	21,000	4	1	4	960×960	0.0001	0.582	1.5
U-Net	DDR	83,600	4	1	4	960×960	0.0001	0.556	5.4
DeepLabv3	IDRiD	6,000	1	16	16	960×960	0.007	0.428	3.6
DeepLabv3	DDR	21,200	1	16	16	960×960	0.007	0.436	12.8
HRNet-W48	IDRiD	7,000	2	6	12	960×960	0.01	0.618	2.6
HRNet-W48	DDR	28,000	2	6	12	960×960	0.01	0.617	10.1
Swin-Unet	IDRiD	63,000	24	1	24	224×224	0.01	0.644	2.6
Swin-Unet	DDR	255,600	24	1	24	224×224	0.01	0.620	10.4
FCT	IDRiD	129,000	4	1	4	384×384	0.001	0.257	19.7
FCT	DDR	521,600	4	1	4	384×384	0.001	0.175	72.0
nnU-Net	IDRiD	250,000	2	1	2	1024×1536	0.01	–[n]	7.2
nnU-Net	DDR	250,000	2	1	2	1024×1536	0.01	–[n]	7.2
nnU-Net ResEnc-M	IDRiD	250,000	13	1	13	384×640	0.01	–[n]	12.8
nnU-Net ResEnc-M	DDR	250,000	12	1	12	512×512	0.01	–[n]	11.4
M2MRF	IDRiD	21,000	4	1	4	960×960	0.01	0.541	3.0
M2MRF	DDR	83,600	4	1	4	960×960	0.01	0.558	11.6
HACDR-Net	IDRiD	83,000	1	1	1	960×960	6e-05	0.518	5.9
HACDR-Net	DDR	334,000	1	1	1	960×960	6e-05	0.383	27.8
HRDecoder	IDRiD	21,000	4	1	4	960×960	0.01	0.562	1.5
HRDecoder	DDR	83,600	4	1	4	960×960	0.01	0.539	5.9
PMCNet	IDRiD	83,000	1	1	1	960×960	0.0001	0.544	1.8
PMCNet	DDR	334,000	1	1	1	960×960	0.0001	0.512	7.1
WSRFNet	IDRiD	21,000	4	1	4	960×960	0.01	0.511	2.7
WSRFNet	DDR	83,600	4	1	4	960×960	0.01	0.524	10.5
CLC-Net	IDRiD	73,000	16	1	16	512×512	0.0003	0.580	9.9
CLC-Net	DDR	293,600	16	1	16	512×512	0.0003	0.539	39.2
TC-Net	IDRiD	73,000	4	1	4	512×512	0.0005	0.570	6.0
TC-Net	DDR	293,600	4	1	4	512×512	0.0005	0.472	24.7
SAMed	IDRiD	13,000	24	1	24	512×512	0.005	0.637	2.1
SAMed	DDR	49,200	24	1	24	512×512	0.005	0.632	7.4
SAM-Adapter	IDRiD	37,000	2	1	2	1024×1024	0.0002	0.617	6.5
SAM-Adapter	DDR	146,800	2	1	2	1024×1024	0.0002	0.589	26.3
SAM2-UNet	IDRiD	7,000	4	3	12	960×960	0.001	0.629	4.1
SAM2-UNet	DDR	28,000	4	3	12	960×960	0.001	0.540	16.2
MedSAM (full FT)	IDRiD	37,000	2	1	2	1024×1024	0.0001	0.631	2.8
MedSAM (full FT)	DDR	146,800	2	1	2	1024×1024	0.0001	0.509	11.1
RETFound + UPerNet	IDRiD	63,000	24	1	24	224×224	0.00046875	0.645	6.1
RETFound + UPerNet	DDR	255,600	24	1	24	224×224	0.00046875	0.643	24.2

Every run's full recipe.json (deviations from the source recipe included) is in the repository's run records. nnU-Net: patch and batch from its own plan for each dataset (runs/NN_*/console.log), 1,000 epochs of 250 steps, SGD at 0.01 with polynomial decay; [n] its checkpoint is nnU-Net's own (the final checkpoint of its schedule), so no validation mAUPR maximum applies.

Supplementary Table 4 Per-class AUPR of every main-table model on every test set (seed 0), with 95% bootstrap intervals. Exact pooled AUPR inside the field of view; the interval is the percentile interval of 2,000 bootstrap replicates over test images. The interval of mAUPR averages the four class values of the same replicates and drops a replicate in which any class has no positive pixel. Both training sources; the internal test is the row whose test set is the training source.

Model	Test set	MA	95% CI	HE	95% CI	EX	95% CI	SE	95% CI	mAUPR	95% CI
Trained on IDRiD											
U-Net	IDRiD	.406	.33–.44	.622	.57–.67	.896	.83–.92	.707	.45–.74	.658	.59–.67
U-Net	DDR	.110	.08–.14	.377	.31–.41	.477	.39–.53	.199	.05–.23	.291	.23–.30
U-Net	TJDR std	.255	.13–.30	.626	.51–.71	.560	.43–.65	.235	.05–.31	.419	.35–.43
U-Net	TJDR UWF	.079	.04–.13	.411	.24–.53	.164	.08–.27	.180	.07–.31	.208	.15–.26
U-Net	MAPLES	.264	.16–.29	.328	.16–.48	.617	.32–.73	.035	.00–.12	.311	.21–.35
U-Net	FGADR	.010	.01–.01	.200	.16–.24	.255	.21–.29	.107	.07–.13	.143	.12–.16
U-Net	Retinal-L.	.044	.03–.04	.186	.13–.18	.208	.13–.21	.104	.06–.12	.136	.10–.13
U-Net	DiaRetDB1	.036	.02–.06	.269	.17–.33	.285	.18–.28	.112	.03–.19	.176	.12–.19
DeepLabv3	IDRiD	.090	.07–.12	.643	.54–.69	.734	.62–.79	.662	.49–.76	.532	.48–.56
DeepLabv3	DDR	.042	.03–.06	.327	.26–.40	.511	.43–.59	.149	.07–.24	.257	.22–.29
DeepLabv3	TJDR std	.133	.07–.21	.605	.49–.69	.585	.32–.75	.307	.12–.52	.407	.33–.48
DeepLabv3	TJDR UWF	.058	.02–.14	.517	.31–.63	.320	.18–.47	.213	.10–.39	.277	.21–.35
DeepLabv3	MAPLES	.160	.12–.21	.237	.08–.43	.661	.41–.77	.087	.01–.21	.286	.20–.35
DeepLabv3	FGADR	.016	.01–.02	.334	.30–.37	.335	.27–.40	.183	.14–.23	.217	.20–.24
DeepLabv3	Retinal-L.	.093	.08–.10	.349	.31–.38	.406	.31–.50	.195	.15–.24	.261	.23–.29
DeepLabv3	DiaRetDB1	.010	.01–.02	.510	.41–.59	.559	.46–.61	.209	.03–.41	.322	.25–.38
HRNet-W48	IDRiD	.446	.41–.48	.684	.63–.72	.884	.82–.91	.783	.59–.87	.699	.64–.73
HRNet-W48	DDR	.146	.11–.19	.438	.38–.51	.507	.42–.58	.254	.12–.38	.336	.29–.39
HRNet-W48	TJDR std	.323	.20–.42	.706	.60–.79	.561	.40–.69	.283	.11–.52	.468	.40–.54
HRNet-W48	TJDR UWF	.111	.05–.20	.480	.28–.64	.132	.07–.25	.296	.13–.49	.255	.18–.33
HRNet-W48	MAPLES	.394	.32–.47	.492	.30–.62	.646	.38–.78	.078	.01–.24	.403	.31–.46
HRNet-W48	FGADR	.025	.02–.03	.375	.33–.42	.362	.30–.42	.234	.19–.29	.249	.22–.27
HRNet-W48	Retinal-L.	.070	.06–.08	.259	.22–.30	.262	.20–.32	.198	.14–.25	.197	.17–.22
HRNet-W48	DiaRetDB1	.039	.02–.06	.456	.36–.54	.382	.27–.45	.212	.06–.34	.272	.21–.31
Swin-Unet	IDRiD	.478	.44–.51	.709	.65–.75	.894	.81–.93	.685	.42–.82	.692	.62–.73
Swin-Unet	DDR	.132	.10–.17	.400	.33–.48	.559	.49–.62	.216	.10–.34	.327	.28–.37
Swin-Unet	TJDR std	.341	.23–.43	.495	.38–.59	.711	.60–.80	.347	.16–.54	.474	.41–.54
Swin-Unet	TJDR UWF	.090	.04–.16	.471	.29–.57	.145	.07–.27	.189	.07–.35	.224	.16–.29
Swin-Unet	MAPLES	.378	.31–.44	.454	.26–.60	.662	.38–.79	.120	.03–.24	.404	.31–.46
Swin-Unet	FGADR	.018	.01–.02	.300	.26–.34	.303	.25–.35	.173	.12–.23	.198	.18–.22
Swin-Unet	Retinal-L.	.083	.07–.09	.289	.24–.32	.359	.27–.42	.197	.15–.24	.232	.20–.25
Swin-Unet	DiaRetDB1	.058	.03–.08	.459	.37–.53	.458	.34–.51	.237	.07–.38	.303	.24–.34
FCT	IDRiD	.077	.06–.10	.163	.10–.23	.684	.57–.74	.110	.03–.26	.259	.22–.30
FCT	DDR	.008	.00–.01	.067	.05–.09	.224	.13–.32	.004	.00–.01	.076	.05–.10
FCT	TJDR std	.000	.00–.00	.006	.00–.01	.004	.00–.01	.004	.00–.01	.004	.00–.01
FCT	TJDR UWF	.001	.00–.00	.020	.01–.04	.021	.01–.05	.006	.00–.01	.012	.01–.02
FCT	MAPLES	.020	.01–.03	.018	.00–.05	.465	.15–.64	.003	.00–.01	.126	.04–.17
FCT	FGADR	.005	.00–.01	.061	.05–.07	.112	.08–.15	.009	.01–.01	.046	.04–.06
FCT	Retinal-L.	.011	.01–.01	.037	.03–.04	.049	.03–.07	.007	.01–.01	.026	.02–.03
FCT	DiaRetDB1	.004	.00–.01	.102	.04–.17	.275	.15–.37	.002	.00–.01	.096	.06–.13
nnU-Net	IDRiD	.499	.42–.51	.671	.61–.72	.876	.79–.91	.641	.40–.79	.671	.60–.70
nnU-Net	DDR	.122	.08–.14	.384	.24–.45	.481	.37–.54	.200	.09–.27	.297	.23–.32
nnU-Net	TJDR std	.210	.13–.25	.615	.48–.70	.490	.32–.60	.421	.21–.64	.434	.36–.49
nnU-Net	TJDR UWF	.088	.03–.15	.419	.20–.59	.169	.08–.25	.144	.06–.26	.205	.13–.26
nnU-Net	MAPLES	.314	.19–.29	.455	.26–.56	.682	.37–.78	.151	.05–.28	.401	.28–.42
nnU-Net	FGADR	.018	.01–.02	.370	.29–.38	.360	.29–.38	.201	.14–.24	.237	.20–.24

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Model	Test set	MA	95% CI	HE	95% CI	EX	95% CI	SE	95% CI	mAUPR	95% CI
nnU-Net	Retinal-L.	.055	.04–.04	.283	.19–.23	.289	.18–.27	.146	.09–.15	.193	.13–.16
nnU-Net	DiaRetDB1	.035	.02–.05	.437	.30–.45	.428	.26–.36	.171	.05–.29	.268	.18–.26
nnU-Net ResEnc-M	IDRiD	.449	.36–.44	.663	.58–.68	.878	.79–.90	.533	.29–.68	.631	.54–.65
nnU-Net ResEnc-M	DDR	.113	.08–.12	.337	.21–.38	.476	.36–.53	.165	.07–.24	.273	.21–.29
nnU-Net ResEnc-M	TJDR std	.217	.12–.25	.555	.41–.61	.526	.36–.61	.385	.15–.58	.421	.33–.46
nnU-Net ResEnc-M	TJDR UWF	.071	.02–.13	.421	.18–.60	.134	.05–.22	.143	.06–.26	.192	.12–.24
nnU-Net ResEnc-M	MAPLES	.269	.17–.26	.472	.25–.57	.677	.34–.74	.017	.00–.08	.359	.24–.36
nnU-Net ResEnc-M	FGADR	.017	.01–.02	.322	.24–.33	.332	.26–.36	.148	.10–.18	.205	.16–.21
nnU-Net ResEnc-M	Retinal-L.	.043	.03–.04	.192	.14–.17	.227	.15–.23	.086	.05–.11	.137	.10–.13
nnU-Net ResEnc-M	DiaRetDB1	.035	.02–.04	.319	.20–.33	.321	.21–.29	.232	.05–.31	.227	.14–.22
M2MRF	IDRiD	.443	.38–.46	.615	.55–.64	.868	.77–.89	.675	.47–.76	.650	.58–.66
M2MRF	DDR	.106	.08–.13	.294	.20–.36	.410	.31–.46	.127	.06–.20	.234	.18–.26
M2MRF	TJDR std	.259	.17–.31	.482	.33–.56	.608	.49–.69	.286	.07–.48	.409	.33–.45
M2MRF	TJDR UWF	.038	.01–.07	.360	.21–.44	.035	.01–.10	.205	.08–.29	.160	.11–.18
M2MRF	MAPLES	.253	.18–.29	.400	.21–.51	.547	.32–.65	.044	.00–.10	.311	.22–.34
M2MRF	FGADR	.007	.01–.01	.082	.06–.09	.172	.11–.22	.083	.05–.11	.086	.06–.10
M2MRF	Retinal-L.	.044	.04–.05	.172	.12–.17	.185	.11–.21	.089	.05–.11	.122	.09–.12
M2MRF	DiaRetDB1	.038	.02–.05	.354	.23–.38	.233	.16–.23	.165	.03–.28	.198	.13–.21
HACDR-Net	IDRiD	.359	.31–.40	.502	.41–.61	.827	.74–.88	.560	.34–.68	.562	.50–.60
HACDR-Net	DDR	.091	.05–.14	.244	.17–.31	.266	.15–.45	.060	.01–.17	.165	.12–.23
HACDR-Net	TJDR std	.378	.19–.52	.475	.33–.59	.641	.55–.72	.529	.16–.76	.506	.39–.59
HACDR-Net	TJDR UWF	.071	.03–.19	.267	.15–.39	.103	.03–.23	.058	.01–.16	.125	.09–.19
HACDR-Net	MAPLES	.280	.21–.35	.177	.04–.41	.579	.27–.73	.035	.00–.15	.268	.18–.34
HACDR-Net	FGADR	.015	.01–.02	.069	.05–.09	.044	.03–.06	.008	.00–.02	.034	.03–.04
HACDR-Net	Retinal-L.	.040	.03–.04	.064	.04–.08	.183	.12–.21	.022	.01–.04	.077	.06–.09
HACDR-Net	DiaRetDB1	.044	.02–.07	.290	.17–.42	.383	.22–.43	.072	.01–.17	.197	.13–.24
HRDecoder	IDRiD	.410	.36–.43	.629	.56–.66	.823	.70–.85	.728	.48–.81	.647	.56–.66
HRDecoder	DDR	.110	.08–.13	.388	.30–.44	.530	.44–.58	.168	.06–.22	.299	.24–.32
HRDecoder	TJDR std	.236	.14–.29	.423	.26–.54	.593	.43–.67	.318	.13–.46	.393	.32–.44
HRDecoder	TJDR UWF	.107	.04–.18	.362	.15–.46	.157	.08–.23	.131	.05–.31	.189	.12–.25
HRDecoder	MAPLES	.326	.24–.37	.464	.26–.59	.597	.33–.70	.121	.03–.23	.377	.28–.41
HRDecoder	FGADR	.020	.02–.02	.323	.27–.35	.293	.22–.34	.179	.13–.22	.204	.17–.22
HRDecoder	Retinal-L.	.053	.04–.05	.180	.14–.18	.182	.12–.20	.116	.07–.14	.133	.10–.13
HRDecoder	DiaRetDB1	.052	.03–.07	.340	.24–.37	.218	.15–.23	.217	.06–.33	.207	.15–.23
PMCNet	IDRiD	.418	.36–.47	.610	.56–.67	.883	.80–.92	.698	.41–.82	.652	.57–.70
PMCNet	DDR	.133	.10–.17	.324	.22–.43	.464	.34–.57	.135	.04–.28	.264	.20–.33
PMCNet	TJDR std	.372	.22–.48	.600	.44–.71	.506	.39–.62	.248	.05–.43	.432	.36–.48
PMCNet	TJDR UWF	.125	.04–.21	.431	.23–.58	.210	.11–.35	.254	.13–.40	.255	.19–.32
PMCNet	MAPLES	.344	.25–.43	.334	.14–.50	.616	.35–.75	.052	.00–.16	.337	.25–.39
PMCNet	FGADR	.023	.02–.03	.335	.29–.38	.355	.31–.40	.133	.09–.18	.212	.19–.23
PMCNet	Retinal-L.	.076	.06–.07	.213	.16–.24	.305	.21–.32	.129	.08–.16	.181	.14–.19
PMCNet	DiaRetDB1	.033	.02–.05	.325	.22–.41	.396	.27–.40	.153	.01–.27	.227	.15–.25
WSRFNet	IDRiD	.388	.33–.40	.588	.51–.61	.849	.70–.88	.502	.18–.65	.582	.47–.61
WSRFNet	DDR	.087	.06–.10	.287	.19–.34	.473	.38–.49	.186	.09–.27	.258	.20–.28
WSRFNet	TJDR std	.198	.12–.24	.426	.27–.51	.539	.40–.59	.220	.06–.43	.346	.27–.39
WSRFNet	TJDR UWF	.061	.03–.09	.379	.22–.48	.116	.04–.20	.230	.09–.32	.196	.13–.23
WSRFNet	MAPLES	.245	.18–.27	.411	.22–.51	.544	.29–.62	.032	.00–.10	.308	.22–.33
WSRFNet	FGADR	.010	.01–.01	.300	.25–.31	.314	.25–.34	.162	.11–.19	.197	.16–.20
WSRFNet	Retinal-L.	.043	.04–.04	.141	.11–.15	.172	.11–.20	.148	.10–.16	.126	.10–.13
WSRFNet	DiaRetDB1	.038	.02–.05	.264	.19–.31	.221	.16–.23	.120	.02–.20	.161	.11–.18
CLC-Net	IDRiD	.437	.37–.46	.683	.59–.71	.889	.81–.92	.634	.40–.73	.661	.58–.68
CLC-Net	DDR	.115	.08–.14	.439	.36–.48	.539	.45–.57	.154	.06–.23	.312	.27–.33

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Model	Test set	MA	95% CI	HE	95% CI	EX	95% CI	SE	95% CI	mAUPR	95% CI
CLC-Net	TJDR std	.295	.16–.38	.631	.50–.71	.618	.45–.73	.292	.12–.43	.459	.38–.51
CLC-Net	TJDR UWF	.091	.03–.16	.466	.25–.57	.170	.08–.26	.105	.04–.18	.208	.14–.25
CLC-Net	MAPLES	.336	.23–.37	.477	.28–.59	.689	.41–.78	.065	.01–.16	.392	.30–.42
CLC-Net	FGADR	.007	.01–.01	.318	.27–.35	.288	.24–.31	.086	.05–.11	.175	.15–.18
CLC-Net	Retinal-L.	.060	.04–.05	.246	.19–.26	.348	.20–.32	.105	.07–.12	.190	.14–.18
CLC-Net	DiaRetDB1	.044	.02–.06	.400	.28–.48	.444	.25–.36	.140	.03–.23	.257	.17–.25
TC-Net	IDRiD	.355	.31–.40	.650	.59–.70	.854	.78–.89	.754	.57–.85	.653	.60–.68
TC-Net	DDR	.119	.09–.15	.404	.32–.49	.572	.50–.63	.153	.07–.25	.312	.28–.35
TC-Net	TJDR std	.233	.16–.31	.605	.50–.69	.691	.58–.77	.398	.15–.61	.482	.41–.54
TC-Net	TJDR UWF	.124	.07–.22	.489	.27–.61	.017	.01–.04	.223	.12–.34	.213	.16–.26
TC-Net	MAPLES	.372	.29–.46	.453	.27–.58	.682	.39–.81	.143	.03–.30	.412	.33–.47
TC-Net	FGADR	.022	.02–.02	.366	.32–.41	.318	.26–.37	.144	.11–.19	.212	.19–.23
TC-Net	Retinal-L.	.080	.07–.08	.270	.22–.30	.237	.16–.29	.114	.08–.15	.175	.15–.19
TC-Net	DiaRetDB1	.039	.02–.06	.433	.34–.51	.426	.29–.48	.218	.09–.32	.279	.21–.31
SAMed	IDRiD	.449	.40–.50	.736	.67–.78	.892	.82–.92	.790	.61–.88	.717	.67–.75
SAMed	DDR	.128	.10–.16	.443	.37–.52	.609	.54–.67	.339	.18–.48	.380	.33–.43
SAMed	TJDR std	.301	.20–.38	.669	.53–.76	.706	.61–.79	.559	.25–.74	.559	.47–.62
SAMed	TJDR UWF	.115	.04–.19	.470	.24–.66	.226	.13–.36	.158	.05–.33	.242	.17–.33
SAMed	MAPLES	.351	.26–.44	.471	.30–.60	.684	.40–.80	.136	.04–.30	.410	.32–.48
SAMed	FGADR	.025	.02–.03	.418	.38–.46	.409	.36–.46	.282	.23–.34	.283	.26–.31
SAMed	Retinal-L.	.083	.07–.09	.354	.31–.38	.362	.28–.42	.219	.16–.26	.255	.22–.27
SAMed	DiaRetDB1	.055	.03–.08	.473	.39–.54	.455	.35–.50	.276	.08–.45	.315	.25–.36
SAM-Adapter	IDRiD	.451	.39–.46	.740	.66–.78	.882	.79–.92	.750	.54–.82	.706	.63–.72
SAM-Adapter	DDR	.117	.08–.14	.403	.30–.46	.560	.47–.59	.157	.08–.23	.309	.26–.33
SAM-Adapter	TJDR std	.256	.14–.33	.581	.43–.69	.563	.40–.63	.351	.16–.55	.438	.36–.49
SAM-Adapter	TJDR UWF	.067	.02–.10	.447	.21–.58	.125	.05–.21	.196	.09–.32	.209	.13–.27
SAM-Adapter	MAPLES	.292	.19–.32	.395	.21–.50	.605	.31–.71	.144	.05–.27	.359	.26–.39
SAM-Adapter	FGADR	.018	.01–.02	.355	.29–.39	.343	.27–.39	.166	.12–.22	.221	.19–.24
SAM-Adapter	Retinal-L.	.046	.04–.04	.246	.17–.23	.317	.17–.32	.142	.10–.16	.188	.13–.18
SAM-Adapter	DiaRetDB1	.037	.02–.06	.449	.30–.49	.365	.19–.32	.130	.03–.26	.245	.16–.25
SAM2-UNet	IDRiD	.439	.37–.45	.670	.59–.72	.871	.78–.90	.678	.43–.77	.664	.58–.69
SAM2-UNet	DDR	.112	.08–.13	.448	.38–.50	.516	.42–.56	.239	.05–.17	.329	.25–.31
SAM2-UNet	TJDR std	.289	.16–.35	.680	.58–.75	.643	.53–.69	.510	.14–.52	.530	.42–.53
SAM2-UNet	TJDR UWF	.091	.03–.14	.482	.30–.62	.121	.04–.23	.230	.06–.31	.231	.15–.26
SAM2-UNet	MAPLES	.319	.23–.37	.404	.21–.53	.644	.38–.75	.097	.01–.21	.366	.27–.40
SAM2-UNet	FGADR	.016	.01–.02	.378	.32–.40	.377	.30–.40	.242	.16–.24	.253	.21–.25
SAM2-UNet	Retinal-L.	.063	.05–.06	.249	.18–.25	.274	.18–.29	.159	.08–.15	.186	.13–.17
SAM2-UNet	DiaRetDB1	.037	.02–.05	.327	.24–.39	.319	.23–.30	.197	.04–.30	.220	.16–.23
MedSAM (full FT)	IDRiD	.373	.33–.41	.665	.57–.73	.855	.77–.90	.674	.43–.83	.642	.57–.69
MedSAM (full FT)	DDR	.144	.11–.18	.400	.35–.46	.555	.49–.62	.155	.08–.25	.314	.28–.35
MedSAM (full FT)	TJDR std	.224	.17–.27	.593	.42–.71	.600	.47–.70	.395	.19–.57	.453	.38–.52
MedSAM (full FT)	TJDR UWF	.119	.05–.24	.443	.24–.57	.213	.12–.34	.403	.25–.55	.295	.22–.37
MedSAM (full FT)	MAPLES	.331	.27–.39	.364	.17–.52	.650	.38–.77	.132	.05–.25	.369	.28–.42
MedSAM (full FT)	FGADR	.024	.02–.03	.367	.33–.41	.394	.35–.44	.254	.20–.31	.260	.24–.28
MedSAM (full FT)	Retinal-L.	.078	.07–.08	.260	.22–.29	.394	.33–.45	.125	.10–.15	.214	.19–.23
MedSAM (full FT)	DiaRetDB1	.043	.02–.07	.417	.30–.52	.535	.42–.60	.131	.03–.24	.282	.22–.32
RETFound + UPerNet	IDRiD	.484	.45–.52	.718	.66–.75	.895	.82–.93	.734	.50–.85	.708	.64–.74
RETFound + UPerNet	DDR	.152	.12–.19	.413	.32–.50	.572	.50–.63	.278	.13–.43	.354	.30–.41
RETFound + UPerNet	TJDR std	.393	.25–.49	.582	.46–.67	.776	.68–.86	.070	.02–.15	.455	.40–.50
RETFound + UPerNet	TJDR UWF	.099	.04–.18	.494	.28–.63	.106	.05–.27	.311	.15–.51	.253	.19–.33
RETFound + UPerNet	MAPLES	.375	.29–.45	.503	.30–.64	.680	.38–.81	.092	.03–.23	.413	.32–.47
RETFound + UPerNet	FGADR	.023	.02–.03	.332	.29–.38	.357	.29–.42	.215	.16–.28	.232	.20–.26

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Model	Test set	MA	95% CI	HE	95% CI	EX	95% CI	SE	95% CI	mAUPR	95% CI
RETFound + UPerNet	Retinal-L.	.081	.06–.08	.288	.23–.32	.299	.22–.36	.156	.11–.20	.206	.17–.22
RETFound + UPerNet	DiaRetDB1	.039	.02–.06	.459	.34–.56	.474	.29–.55	.257	.07–.40	.307	.22–.35
Trained on DDR											
U-Net	IDRiD	.364	.29–.36	.541	.43–.57	.800	.66–.78	.483	.26–.56	.547	.45–.54
U-Net	DDR	.141	.09–.16	.410	.28–.48	.553	.42–.59	.410	.23–.49	.378	.29–.40
U-Net	TJDR std	.296	.16–.34	.676	.55–.75	.614	.42–.65	.392	.13–.49	.494	.39–.50
U-Net	TJDR UWF	.077	.03–.21	.262	.11–.39	.254	.15–.32	.067	.01–.13	.165	.11–.20
U-Net	MAPLES	.222	.13–.25	.462	.24–.58	.509	.24–.50	.125	.01–.21	.329	.22–.33
U-Net	FGADR	.013	.01–.01	.214	.18–.24	.292	.23–.33	.162	.11–.19	.170	.14–.18
U-Net	Retinal-L.	.039	.03–.04	.165	.10–.16	.167	.10–.16	.197	.13–.21	.142	.10–.13
U-Net	DiaRetDB1	.039	.02–.06	.314	.19–.34	.268	.12–.20	.253	.11–.31	.219	.13–.20
DeepLabv3	IDRiD	.052	.04–.07	.640	.49–.72	.708	.58–.77	.630	.40–.75	.507	.43–.55
DeepLabv3	DDR	.036	.02–.05	.447	.30–.61	.614	.54–.68	.387	.25–.50	.371	.31–.42
DeepLabv3	TJDR std	.172	.08–.28	.649	.51–.75	.724	.64–.76	.411	.15–.61	.489	.42–.54
DeepLabv3	TJDR UWF	.071	.03–.10	.066	.02–.20	.296	.17–.44	.058	.02–.18	.123	.08–.19
DeepLabv3	MAPLES	.143	.10–.19	.452	.24–.58	.641	.41–.73	.108	.01–.24	.336	.25–.39
DeepLabv3	FGADR	.011	.01–.01	.339	.30–.38	.356	.30–.41	.224	.18–.27	.232	.21–.25
DeepLabv3	Retinal-L.	.074	.06–.08	.316	.25–.37	.440	.32–.52	.253	.19–.31	.271	.22–.30
DeepLabv3	DiaRetDB1	.008	.00–.01	.495	.40–.57	.545	.44–.59	.241	.03–.40	.322	.26–.36
HRNet-W48	IDRiD	.372	.34–.40	.694	.58–.76	.834	.75–.88	.679	.42–.79	.645	.57–.68
HRNet-W48	DDR	.155	.12–.20	.477	.39–.59	.590	.52–.67	.550	.42–.65	.443	.39–.49
HRNet-W48	TJDR std	.412	.24–.53	.788	.71–.85	.713	.63–.76	.581	.25–.76	.624	.54–.67
HRNet-W48	TJDR UWF	.116	.06–.24	.120	.04–.29	.248	.15–.36	.107	.02–.27	.148	.10–.22
HRNet-W48	MAPLES	.383	.31–.45	.500	.28–.63	.701	.48–.79	.220	.05–.37	.451	.37–.50
HRNet-W48	FGADR	.026	.02–.03	.399	.35–.45	.394	.34–.45	.255	.20–.32	.269	.24–.30
HRNet-W48	Retinal-L.	.083	.07–.09	.251	.20–.31	.339	.27–.40	.332	.28–.38	.251	.22–.28
HRNet-W48	DiaRetDB1	.034	.02–.05	.437	.34–.53	.432	.32–.52	.362	.21–.46	.316	.26–.35
Swin-Unet	IDRiD	.420	.38–.46	.625	.52–.69	.853	.76–.89	.615	.41–.73	.628	.56–.67
Swin-Unet	DDR	.174	.13–.22	.479	.39–.58	.650	.59–.71	.424	.29–.53	.432	.39–.48
Swin-Unet	TJDR std	.370	.24–.45	.676	.56–.76	.696	.53–.79	.438	.13–.63	.545	.46–.60
Swin-Unet	TJDR UWF	.107	.05–.22	.336	.19–.47	.271	.18–.37	.044	.01–.14	.189	.14–.25
Swin-Unet	MAPLES	.395	.32–.46	.509	.31–.64	.726	.50–.81	.200	.06–.35	.457	.38–.51
Swin-Unet	FGADR	.021	.02–.02	.349	.31–.39	.382	.33–.44	.216	.17–.27	.242	.22–.27
Swin-Unet	Retinal-L.	.113	.09–.12	.254	.20–.30	.418	.33–.48	.242	.19–.29	.257	.22–.28
Swin-Unet	DiaRetDB1	.048	.03–.07	.381	.28–.47	.496	.40–.55	.255	.14–.36	.295	.25–.32
FCT	IDRiD	.005	.00–.01	.153	.10–.21	.679	.56–.74	.046	.01–.13	.221	.20–.24
FCT	DDR	.001	.00–.00	.163	.11–.22	.524	.46–.58	.014	.01–.02	.176	.15–.19
FCT	TJDR std	.001	.00–.00	.099	.05–.15	.596	.43–.68	.040	.01–.11	.184	.15–.20
FCT	TJDR UWF	.001	.00–.00	.055	.02–.10	.135	.06–.24	.009	.00–.02	.050	.03–.08
FCT	MAPLES	.004	.00–.01	.030	.01–.08	.510	.21–.66	.007	.00–.02	.138	.06–.18
FCT	FGADR	.004	.00–.00	.065	.05–.08	.182	.14–.23	.015	.01–.02	.067	.06–.08
FCT	Retinal-L.	.010	.01–.01	.057	.04–.07	.210	.14–.28	.018	.01–.02	.074	.05–.09
FCT	DiaRetDB1	.000	.00–.00	.087	.02–.17	.270	.16–.34	.009	.00–.02	.092	.05–.12
nnU-Net	IDRiD	.334	.24–.34	.637	.52–.71	.810	.71–.86	.623	.38–.71	.601	.51–.63
nnU-Net	DDR	.154	.11–.17	.501	.39–.57	.614	.51–.66	.458	.30–.56	.432	.36–.46
nnU-Net	TJDR std	.309	.17–.36	.705	.61–.75	.635	.45–.71	.425	.15–.61	.518	.42–.55
nnU-Net	TJDR UWF	.111	.03–.21	.374	.18–.54	.157	.07–.26	.142	.06–.27	.196	.13–.26
nnU-Net	MAPLES	.372	.18–.29	.493	.27–.58	.717	.38–.75	.196	.06–.29	.445	.29–.42
nnU-Net	FGADR	.020	.01–.02	.396	.33–.41	.389	.31–.42	.232	.18–.27	.259	.22–.27
nnU-Net	Retinal-L.	.067	.04–.05	.232	.14–.19	.319	.17–.28	.224	.17–.23	.211	.14–.18
nnU-Net	DiaRetDB1	.065	.04–.09	.462	.31–.48	.476	.26–.39	.295	.10–.35	.325	.21–.30
nnU-Net ResEnc-M	IDRiD	.349	.26–.32	.664	.54–.72	.832	.73–.87	.603	.36–.67	.612	.51–.62

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Model	Test set	MA	95% CI	HE	95% CI	EX	95% CI	SE	95% CI	mAUPR	95% CI
nnU-Net ResEnc-M	DDR	.147	.10–.16	.481	.37–.55	.653	.56–.70	.432	.28–.52	.428	.36–.45
nnU-Net ResEnc-M	TJDR std	.320	.19–.34	.751	.63–.81	.673	.48–.75	.538	.19–.69	.571	.46–.60
nnU-Net ResEnc-M	TJDR UWF	.115	.03–.21	.317	.17–.44	.132	.06–.24	.112	.02–.23	.169	.11–.22
nnU-Net ResEnc-M	MAPLES	.341	.16–.29	.469	.26–.58	.762	.42–.78	.293	.12–.37	.466	.32–.45
nnU-Net ResEnc-M	FGADR	.018	.01–.02	.409	.33–.41	.401	.31–.42	.242	.17–.27	.268	.22–.27
nnU-Net ResEnc-M	Retinal-L.	.062	.03–.04	.269	.13–.21	.355	.20–.29	.257	.18–.24	.236	.15–.18
nnU-Net ResEnc-M	DiaRetDB1	.062	.03–.08	.450	.28–.44	.475	.27–.37	.261	.06–.32	.312	.20–.28
M2MRF	IDRiD	.392	.34–.40	.555	.45–.56	.776	.68–.78	.563	.35–.66	.571	.49–.57
M2MRF	DDR	.148	.10–.18	.402	.27–.48	.547	.46–.59	.422	.25–.49	.380	.30–.40
M2MRF	TJDR std	.299	.19–.35	.696	.58–.75	.618	.50–.68	.412	.14–.55	.506	.42–.53
M2MRF	TJDR UWF	.093	.04–.17	.403	.22–.51	.209	.11–.30	.037	.01–.10	.186	.13–.22
M2MRF	MAPLES	.266	.17–.30	.437	.24–.55	.454	.30–.50	.138	.02–.25	.324	.24–.35
M2MRF	FGADR	.019	.01–.02	.310	.24–.33	.340	.27–.38	.181	.13–.22	.212	.18–.22
M2MRF	Retinal-L.	.050	.04–.05	.183	.13–.17	.174	.12–.18	.195	.13–.22	.150	.11–.14
M2MRF	DiaRetDB1	.049	.02–.07	.218	.15–.26	.177	.13–.18	.247	.14–.30	.173	.13–.18
HACDR-Net	IDRiD	.225	.18–.26	.365	.28–.47	.789	.67–.83	.199	.04–.44	.395	.34–.45
HACDR-Net	DDR	.114	.08–.16	.309	.22–.40	.485	.37–.58	.168	.08–.28	.269	.22–.32
HACDR-Net	TJDR std	.356	.21–.46	.632	.54–.71	.553	.45–.64	.196	.03–.39	.434	.37–.49
HACDR-Net	TJDR UWF	.095	.03–.20	.385	.20–.50	.204	.11–.32	.130	.04–.24	.203	.14–.27
HACDR-Net	MAPLES	.224	.15–.29	.378	.17–.54	.512	.22–.63	.093	.01–.22	.302	.21–.35
HACDR-Net	FGADR	.021	.02–.02	.186	.15–.22	.134	.10–.17	.049	.03–.07	.098	.08–.11
HACDR-Net	Retinal-L.	.058	.04–.06	.163	.12–.19	.166	.11–.19	.138	.10–.17	.131	.10–.14
HACDR-Net	DiaRetDB1	.016	.01–.03	.228	.12–.32	.319	.15–.34	.101	.02–.19	.166	.10–.20
HRDecoder	IDRiD	.357	.31–.36	.547	.41–.57	.714	.60–.73	.539	.35–.60	.539	.46–.54
HRDecoder	DDR	.124	.08–.16	.318	.20–.40	.496	.40–.56	.245	.11–.33	.296	.23–.32
HRDecoder	TJDR std	.334	.19–.41	.689	.59–.75	.614	.51–.67	.358	.12–.51	.499	.42–.53
HRDecoder	TJDR UWF	.119	.05–.18	.368	.17–.53	.180	.09–.27	.124	.05–.23	.198	.13–.25
HRDecoder	MAPLES	.231	.14–.26	.407	.23–.50	.427	.22–.48	.127	.03–.22	.298	.21–.32
HRDecoder	FGADR	.014	.01–.02	.267	.21–.29	.309	.25–.35	.183	.14–.22	.193	.16–.20
HRDecoder	Retinal-L.	.043	.03–.04	.126	.08–.14	.171	.11–.17	.168	.13–.19	.127	.09–.12
HRDecoder	DiaRetDB1	.045	.02–.06	.275	.17–.32	.234	.13–.19	.203	.08–.29	.189	.13–.19
PMCNet	IDRiD	.348	.31–.38	.458	.37–.51	.758	.68–.81	.524	.20–.71	.522	.43–.57
PMCNet	DDR	.157	.12–.20	.346	.23–.46	.482	.37–.58	.262	.14–.38	.312	.25–.36
PMCNet	TJDR std	.406	.25–.50	.639	.54–.72	.476	.33–.60	.308	.06–.49	.457	.38–.51
PMCNet	TJDR UWF	.085	.03–.19	.427	.23–.55	.217	.12–.32	.091	.03–.19	.205	.14–.26
PMCNet	MAPLES	.371	.28–.44	.444	.24–.59	.627	.34–.74	.131	.00–.28	.393	.30–.43
PMCNet	FGADR	.021	.02–.02	.323	.27–.37	.339	.29–.37	.162	.11–.21	.211	.19–.23
PMCNet	Retinal-L.	.105	.08–.10	.236	.19–.25	.272	.18–.26	.128	.09–.16	.185	.15–.18
PMCNet	DiaRetDB1	.036	.02–.06	.312	.21–.40	.366	.23–.38	.266	.04–.42	.245	.16–.28
WSRFNet	IDRiD	.319	.24–.32	.489	.36–.51	.673	.53–.72	.384	.17–.49	.466	.37–.47
WSRFNet	DDR	.088	.05–.11	.292	.20–.37	.422	.31–.48	.135	.05–.21	.234	.18–.26
WSRFNet	TJDR std	.173	.08–.26	.588	.43–.64	.433	.23–.51	.114	.01–.20	.327	.24–.35
WSRFNet	TJDR UWF	.038	.01–.18	.323	.16–.39	.155	.06–.24	.024	.00–.08	.135	.08–.17
WSRFNet	MAPLES	.264	.17–.29	.428	.21–.53	.436	.26–.43	.104	.01–.18	.308	.22–.31
WSRFNet	FGADR	.011	.01–.01	.158	.12–.17	.086	.05–.09	.020	.01–.03	.069	.05–.07
WSRFNet	Retinal-L.	.037	.03–.04	.085	.05–.10	.107	.07–.12	.062	.03–.09	.073	.05–.08
WSRFNet	DiaRetDB1	.050	.02–.07	.215	.15–.25	.178	.13–.19	.158	.08–.20	.150	.12–.16
CLC-Net	IDRiD	.358	.29–.37	.614	.47–.64	.830	.71–.86	.570	.25–.67	.593	.48–.60
CLC-Net	DDR	.152	.10–.18	.466	.34–.53	.612	.51–.64	.428	.25–.51	.414	.34–.43
CLC-Net	TJDR std	.352	.19–.42	.666	.55–.73	.667	.49–.73	.296	.09–.45	.495	.42–.51
CLC-Net	TJDR UWF	.110	.04–.21	.137	.02–.14	.159	.06–.28	.095	.02–.29	.125	.06–.18
CLC-Net	MAPLES	.287	.17–.31	.475	.26–.56	.611	.33–.61	.030	.00–.11	.351	.24–.34

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Model	Test set	MA	95% CI	HE	95% CI	EX	95% CI	SE	95% CI	mAUPR	95% CI
CLC-Net	FGADR	.011	.01–.01	.304	.24–.32	.305	.22–.34	.089	.05–.11	.177	.14–.18
CLC-Net	Retinal-L.	.048	.04–.05	.164	.08–.16	.222	.12–.22	.173	.11–.20	.152	.10–.14
CLC-Net	DiaRetDB1	.040	.02–.06	.318	.16–.38	.260	.16–.23	.242	.05–.31	.215	.13–.21
TC-Net	IDRiD	.225	.20–.25	.593	.47–.67	.773	.67–.82	.636	.38–.76	.557	.48–.60
TC-Net	DDR	.112	.08–.14	.476	.40–.54	.607	.55–.65	.250	.12–.39	.361	.32–.40
TC-Net	TJDR std	.239	.12–.34	.598	.50–.67	.670	.52–.75	.355	.09–.52	.466	.39–.52
TC-Net	TJDR UWF	.100	.04–.22	.383	.20–.53	.234	.14–.35	.056	.02–.17	.193	.15–.26
TC-Net	MAPLES	.241	.17–.30	.469	.24–.61	.722	.48–.81	.027	.00–.09	.365	.28–.40
TC-Net	FGADR	.018	.02–.02	.334	.30–.37	.349	.29–.40	.156	.12–.20	.214	.19–.23
TC-Net	Retinal-L.	.056	.04–.06	.206	.16–.24	.314	.22–.35	.145	.10–.19	.180	.14–.20
TC-Net	DiaRetDB1	.023	.01–.04	.475	.37–.55	.509	.36–.53	.188	.05–.27	.299	.23–.32
SAMed	IDRiD	.373	.33–.42	.653	.52–.73	.845	.76–.89	.696	.49–.80	.642	.57–.68
SAMed	DDR	.173	.13–.22	.574	.52–.64	.669	.62–.72	.460	.30–.60	.469	.42–.51
SAMed	TJDR std	.428	.29–.51	.743	.65–.81	.707	.57–.79	.570	.26–.73	.612	.53–.66
SAMed	TJDR UWF	.130	.05–.28	.450	.23–.60	.208	.12–.32	.043	.01–.19	.208	.14–.29
SAMed	MAPLES	.365	.28–.44	.515	.30–.65	.732	.51–.82	.227	.07–.36	.460	.38–.51
SAMed	FGADR	.027	.02–.03	.412	.37–.45	.442	.39–.49	.279	.23–.34	.290	.27–.31
SAMed	Retinal-L.	.109	.09–.12	.316	.26–.37	.384	.30–.44	.304	.25–.35	.278	.24–.30
SAMed	DiaRetDB1	.047	.02–.07	.480	.38–.56	.468	.37–.52	.295	.12–.43	.322	.27–.35
SAM-Adapter	IDRiD	.411	.34–.42	.644	.51–.74	.804	.71–.85	.649	.42–.73	.627	.54–.65
SAM-Adapter	DDR	.147	.10–.18	.538	.43–.62	.592	.51–.63	.454	.30–.56	.433	.38–.45
SAM-Adapter	TJDR std	.358	.22–.43	.751	.67–.82	.625	.51–.67	.432	.19–.64	.542	.46–.58
SAM-Adapter	TJDR UWF	.111	.04–.17	.387	.20–.56	.193	.09–.31	.104	.02–.23	.199	.13–.26
SAM-Adapter	MAPLES	.277	.18–.31	.454	.25–.57	.570	.30–.61	.187	.08–.30	.372	.26–.40
SAM-Adapter	FGADR	.019	.02–.02	.384	.32–.41	.358	.29–.41	.219	.17–.26	.245	.21–.26
SAM-Adapter	Retinal-L.	.054	.04–.05	.217	.13–.19	.273	.15–.31	.228	.18–.25	.193	.14–.19
SAM-Adapter	DiaRetDB1	.071	.04–.10	.461	.28–.50	.412	.19–.33	.221	.05–.30	.291	.18–.27
SAM2-UNet	IDRiD	.351	.29–.36	.524	.43–.56	.782	.68–.81	.572	.23–.67	.557	.45–.57
SAM2-UNet	DDR	.153	.10–.18	.493	.36–.55	.518	.41–.56	.278	.11–.29	.360	.28–.36
SAM2-UNet	TJDR std	.325	.18–.37	.647	.55–.71	.640	.51–.69	.416	.11–.49	.507	.40–.52
SAM2-UNet	TJDR UWF	.099	.04–.18	.382	.20–.52	.219	.10–.30	.118	.03–.22	.204	.14–.24
SAM2-UNet	MAPLES	.331	.22–.38	.442	.23–.55	.715	.46–.76	.104	.01–.21	.398	.31–.42
SAM2-UNet	FGADR	.017	.01–.02	.356	.28–.37	.374	.29–.39	.241	.15–.24	.247	.20–.24
SAM2-UNet	Retinal-L.	.084	.06–.08	.215	.13–.20	.322	.21–.29	.301	.21–.31	.231	.16–.20
SAM2-UNet	DiaRetDB1	.040	.02–.06	.331	.23–.35	.409	.27–.35	.286	.08–.44	.267	.18–.27
MedSAM (full FT)	IDRiD	.331	.29–.37	.676	.55–.76	.806	.72–.86	.522	.00–.20	.584	.43–.53
MedSAM (full FT)	DDR	.136	.10–.19	.529	.42–.64	.626	.57–.68	.361	.07–.23	.413	.32–.40
MedSAM (full FT)	TJDR std	.347	.24–.42	.740	.65–.81	.686	.57–.77	.440	.00–.14	.553	.42–.49
MedSAM (full FT)	TJDR UWF	.129	.05–.24	.577	.34–.70	.225	.14–.33	.041	.00–.00	.243	.16–.29
MedSAM (full FT)	MAPLES	.369	.29–.44	.515	.29–.65	.687	.43–.81	.097	.00–.00	.417	.32–.43
MedSAM (full FT)	FGADR	.018	.02–.02	.419	.38–.46	.424	.37–.48	.209	.03–.07	.268	.21–.25
MedSAM (full FT)	Retinal-L.	.097	.08–.10	.359	.29–.39	.421	.33–.48	.174	.01–.05	.262	.19–.24
MedSAM (full FT)	DiaRetDB1	.045	.02–.07	.462	.36–.55	.487	.39–.55	.141	.00–.16	.284	.21–.30
RETFound + UPerNet	IDRiD	.446	.41–.48	.681	.57–.74	.845	.76–.88	.603	.33–.75	.644	.56–.69
RETFound + UPerNet	DDR	.189	.14–.24	.491	.39–.60	.640	.56–.71	.546	.38–.66	.466	.41–.52
RETFound + UPerNet	TJDR std	.387	.28–.46	.733	.65–.79	.724	.56–.81	.513	.17–.70	.589	.50–.64
RETFound + UPerNet	TJDR UWF	.073	.02–.17	.064	.02–.18	.278	.18–.39	.030	.01–.09	.111	.07–.17
RETFound + UPerNet	MAPLES	.420	.35–.47	.531	.30–.67	.722	.49–.81	.166	.03–.31	.460	.37–.50
RETFound + UPerNet	FGADR	.023	.02–.03	.419	.37–.45	.406	.34–.45	.228	.17–.29	.269	.24–.29
RETFound + UPerNet	Retinal-L.	.107	.08–.10	.321	.22–.35	.419	.30–.43	.243	.17–.29	.273	.21–.27
RETFound + UPerNet	DiaRetDB1	.043	.02–.06	.430	.30–.52	.477	.33–.48	.306	.12–.42	.314	.24–.34

TJDR UWF, ultra-widefield images predicted at native resolution with sliding windows. Retinal-L., Retinal-Lesions without its grey-level 127 regions. DiaRetDB1, pixel AUPR against the ≥ 0.75 expert consensus (the official image-level

protocol is Supplementary Table 13).

Supplementary Table 5 Secondary metrics of every main-table model on the internal tests (mean over the four classes and over seeds). 11-thr., the 11-threshold AUPR of the M2MRF evaluation code on the same maps; Dice and IoU at the threshold chosen on the validation split; lesion-level precision and recall by lesion size (S, M, L: small, medium, large; Methods) at the same threshold; image-level AUC of the maximum probability inside the field of view.

Model	Seeds	mAUPR	11-thr.	Dice	IoU	Lesion prec.	Recall S	Recall M	Recall L	Image AUC
Trained on IDRiD (internal test; mean over seeds)										
U-Net	3	.644 ± .015	.638 ± .013	.622 ± .020	.465 ± .021	.489 ± .023	.654 ± .047	.851 ± .032	.800 ± .040	.797 ± .041
DeepLabv3	3	.537 ± .006	.536 ± .004	.515 ± .006	.366 ± .005	.551 ± .019	.270 ± .049	.533 ± .033	.776 ± .026	–
HRNet-W48	3	.700 ± .004	.691 ± .003	.627 ± .004	.468 ± .004	.485 ± .010	.584 ± .040	.879 ± .013	.862 ± .018	–
Swin-Unet	3	.688 ± .003	.686 ± .002	.640 ± .002	.481 ± .002	.419 ± .007	.671 ± .050	.849 ± .008	.827 ± .014	–
FCT	3	.298 ± .084	.359 ± .071	.330 ± .066	.217 ± .048	.267 ± .047	.205 ± .032	.523 ± .029	.726 ± .027	–
nnU-Net	3	.665 ± .008	.675 ± .006	.656 ± .009	.498 ± .010	.552 ± .016	.649 ± .044	.828 ± .003	.739 ± .004	.786 ± .016
nnU-Net ResEnc-M	1	.631	.631	.638	.479	.526	.600	.852	.695	–
M2MRF	3	.657 ± .010	.650 ± .004	.648 ± .004	.490 ± .004	.531 ± .008	.615 ± .015	.824 ± .009	.736 ± .018	–
HACDR-Net	3	.538 ± .031	.559 ± .013	.520 ± .025	.367 ± .022	.495 ± .040	.411 ± .019	.749 ± .046	.764 ± .014	.698 ± .163
HRDecoder	1	.647	.637	.636	.476	.605	.456	.792	.785	.885
PMCNet	1	.652	.648	.616	.458	.529	.592	.825	.804	.786
WSRFNet	1	.582	.605	.641	.482	.538	.611	.808	.701	.852
CLC-Net	1	.661	.665	.632	.475	.495	.565	.832	.864	.868
TC-Net	1	.653	.643	.607	.449	.536	.443	.804	.815	.874
SAMed	3	.706 ± .015	.703 ± .012	.658 ± .007	.502 ± .008	.519 ± .007	.553 ± .055	.798 ± .033	.838 ± .030	.907 ± .033
SAM-Adapter	1	.706	.704	.663	.508	.527	.624	.851	.809	.832
SAM2-UNet	1	.664	.683	.653	.495	.440	.846	.925	.826	.835
MedSAM (full FT)	1	.642	.654	.609	.451	.507	.392	.759	.791	.791
RETFound + UPerNet	3	.699 ± .009	.692 ± .008	.650 ± .008	.492 ± .009	.469 ± .014	.683 ± .015	.873 ± .014	.856 ± .028	.821 ± .040
Trained on DDR (internal test; mean over seeds)										
U-Net	3	.371 ± .009	.421 ± .008	.405 ± .017	.260 ± .014	.435 ± .042	.247 ± .016	.482 ± .012	.646 ± .009	.786 ± .008
DeepLabv3	3	.377 ± .005	.382 ± .003	.380 ± .006	.246 ± .004	.497 ± .015	.062 ± .005	.287 ± .020	.655 ± .010	–
HRNet-W48	3	.447 ± .009	.461 ± .009	.448 ± .004	.297 ± .004	.513 ± .006	.217 ± .007	.495 ± .006	.680 ± .012	–
Swin-Unet	3	.431 ± .001	.447 ± .003	.440 ± .003	.289 ± .003	.424 ± .008	.312 ± .012	.543 ± .013	.674 ± .002	–
FCT	1	.176	.344	.197	.125	.224	.096	.351	.608	–
nnU-Net	3	.431 ± .003	.467 ± .001	.458 ± .010	.304 ± .008	.515 ± .008	.258 ± .006	.470 ± .006	.622 ± .007	.796 ± .004
nnU-Net ResEnc-M	1	.428	.448	.463	.310	.491	.294	.490	.664	–
M2MRF	3	.369 ± .011	.406 ± .008	.407 ± .012	.262 ± .009	.454 ± .007	.276 ± .003	.556 ± .001	.696 ± .023	–
HACDR-Net	3	.252 ± .017	.289 ± .022	.289 ± .022	.175 ± .014	.416 ± .019	.106 ± .022	.350 ± .028	.562 ± .016	.719 ± .012
HRDecoder	1	.296	.333	.373	.236	.459	.234	.496	.629	.714
PMCNet	1	.312	.342	.338	.206	.431	.200	.454	.629	.770
WSRFNet	1	.234	.264	.341	.211	.431	.284	.458	.531	.659
CLC-Net	1	.414	.433	.462	.309	.506	.266	.470	.639	.781
TC-Net	1	.361	.365	.352	.223	.443	.102	.376	.634	.785
SAMed	3	.460 ± .027	.472 ± .027	.454 ± .025	.301 ± .022	.497 ± .008	.242 ± .029	.474 ± .031	.674 ± .007	.842 ± .005
SAM-Adapter	1	.433	.465	.457	.304	.471	.257	.496	.690	.657
SAM2-UNet	1	.360	.445	.413	.265	.459	.298	.559	.698	.749
MedSAM (full FT)	1	.413	.433	.432	.283	.516	.212	.484	.675	.795
RETFound + UPerNet	3	.450 ± .018	.458 ± .016	.447 ± .006	.294 ± .005	.472 ± .006	.290 ± .006	.538 ± .007	.696 ± .009	.844 ± .002

±, standard deviation over seeds where more than one seed was run.

Supplementary Table 6 Registered and revised recipes of the three models whose per-image loss term emptied a rare-class channel (seed 0, internal tests). The registered recipe is the one frozen before training (per-image Dice or IoU with smoothing 1); the revised recipe pools the term over the batch (Methods, rare-class collapse). The main tables carry the revised runs; the registered runs stay in the repository under the `_registered` suffix.

Model	Source	Recipe	MA	HE	EX	SE	mAUPR
U-Net	IDRiD	registered	.382	.541	.858	.644	.606
U-Net	IDRiD	revised (main table)	.406	.622	.896	.707	.658
U-Net	DDR	registered	.006	.010	.460	.000	.119
U-Net	DDR	revised (main table)	.141	.410	.553	.410	.378
SAM2-UNet	IDRiD	registered	.439	.665	.870	.571	.636
SAM2-UNet	IDRiD	revised (main table)	.439	.670	.871	.678	.664
SAM2-UNet	DDR	registered	.023	.265	.528	.022	.210
SAM2-UNet	DDR	revised (main table)	.153	.493	.518	.278	.360
MedSAM	IDRiD	registered	.382	.731	.867	.357	.584
MedSAM	IDRiD	revised (main table)	.373	.665	.855	.674	.642
MedSAM	DDR	registered	.000	.487	.573	.001	.265
MedSAM	DDR	revised (main table)	.136	.529	.626	.361	.413

Supplementary Table 7 Every factor cell, per class (seed 0). One factor is varied at a time on the representative models at the DDR pixel budget (Methods); the variant column names the level, the training-set column the data the cell was trained on. Each cell is listed on every test set it was applied to; in-source tests of merged cells are the rows whose test set was part of the training set.

Model	Level	Trained on	Test set	MA	HE	EX	SE	mAUPR
Training source								
M2MRF	bstd	FGADR	IDRiD	.141	.320	.729	.557	.437
M2MRF	bstd	FGADR	DDR	.055	.219	.404	.107	.197
M2MRF	bstd	FGADR	TJDR std	.267	.642	.666	.223	.449
M2MRF	bstd	FGADR	TJDR UWF	.035	.480	.282	.176	.243
M2MRF	bstd	FGADR	MAPLES	.086	.336	.492	.090	.251
M2MRF	bstd	FGADR	FGADR	.066	.511	.622	.238	.359
M2MRF	bstd	FGADR	Retinal-L.	.031	.179	.177	.167	.139
M2MRF	bstd	FGADR	DiaRetDB1	.006	.270	.287	.142	.176
M2MRF	bstd	TJDR-std	IDRiD	.138	.200	.695	.447	.370
M2MRF	bstd	TJDR-std	DDR	.065	.244	.527	.086	.231
M2MRF	bstd	TJDR-std	TJDR std	.325	.717	.653	.358	.513
M2MRF	bstd	TJDR-std	TJDR UWF	.009	.296	.130	.197	.158
M2MRF	bstd	TJDR-std	MAPLES	.089	.331	.403	.020	.211
M2MRF	bstd	TJDR-std	FGADR	.016	.228	.294	.132	.167
M2MRF	bstd	TJDR-std	Retinal-L.	.024	.089	.173	.077	.091
M2MRF	bstd	TJDR-std	DiaRetDB1	.012	.189	.231	.156	.147
RETFound + UPerNet	bstd	FGADR	IDRiD	.266	.598	.823	.652	.585
RETFound + UPerNet	bstd	FGADR	DDR	.029	.415	.511	.346	.325
RETFound + UPerNet	bstd	FGADR	TJDR std	.195	.746	.593	.403	.484
RETFound + UPerNet	bstd	FGADR	TJDR UWF	.066	.630	.341	.178	.304
RETFound + UPerNet	bstd	FGADR	MAPLES	.207	.508	.735	.147	.399
RETFound + UPerNet	bstd	FGADR	FGADR	.124	.650	.699	.349	.456
RETFound + UPerNet	bstd	FGADR	Retinal-L.	.141	.387	.454	.313	.324
RETFound + UPerNet	bstd	FGADR	DiaRetDB1	.023	.578	.623	.235	.365
RETFound + UPerNet	bstd	TJDR-std	IDRiD	.270	.461	.800	.466	.499
RETFound + UPerNet	bstd	TJDR-std	DDR	.130	.348	.626	.253	.339
RETFound + UPerNet	bstd	TJDR-std	TJDR std	.440	.831	.823	.714	.702
RETFound + UPerNet	bstd	TJDR-std	TJDR UWF	.060	.565	.103	.231	.240
RETFound + UPerNet	bstd	TJDR-std	MAPLES	.270	.361	.542	.060	.308
RETFound + UPerNet	bstd	TJDR-std	FGADR	.024	.344	.365	.183	.229
RETFound + UPerNet	bstd	TJDR-std	Retinal-L.	.078	.210	.308	.144	.185
RETFound + UPerNet	bstd	TJDR-std	DiaRetDB1	.020	.376	.466	.269	.283
SAM2-UNet	bstd	FGADR	IDRiD	.248	.534	.817	.544	.536
SAM2-UNet	bstd	FGADR	DDR	.076	.341	.445	.305	.292
SAM2-UNet	bstd	FGADR	TJDR std	.352	.640	.648	.651	.573
SAM2-UNet	bstd	FGADR	TJDR UWF	.076	.578	.300	.511	.366
SAM2-UNet	bstd	FGADR	MAPLES	.165	.367	.669	.034	.309
SAM2-UNet	bstd	FGADR	FGADR	.096	.527	.640	.357	.405
SAM2-UNet	bstd	FGADR	Retinal-L.	.050	.205	.192	.233	.170
SAM2-UNet	bstd	FGADR	DiaRetDB1	.023	.285	.257	.221	.196
SAM2-UNet	bstd	TJDR-std	IDRiD	.013	.395	.780	.004	.298
SAM2-UNet	bstd	TJDR-std	DDR	.011	.316	.549	.001	.219
SAM2-UNet	bstd	TJDR-std	TJDR std	.024	.758	.702	.026	.378
SAM2-UNet	bstd	TJDR-std	TJDR UWF	.007	.389	.137	.000	.133
SAM2-UNet	bstd	TJDR-std	MAPLES	.005	.275	.431	.000	.178
SAM2-UNet	bstd	TJDR-std	FGADR	.006	.287	.310	.002	.151
SAM2-UNet	bstd	TJDR-std	Retinal-L.	.009	.119	.220	.003	.088

continued on the next page

continued

Model	Level	Trained on	Test set	MA	HE	EX	SE	mAUPR
SAM2-UNet	bstd	TJDR-std	DiaRetDB1	.000	.218	.256	.001	.119
U-Net	bstd	FGADR	IDRiD	.092	.294	.793	.276	.364
U-Net	bstd	FGADR	DDR	.032	.259	.467	.199	.239
U-Net	bstd	FGADR	TJDR std	.135	.718	.603	.339	.449
U-Net	bstd	FGADR	TJDR UWF	.028	.482	.208	.273	.248
U-Net	bstd	FGADR	MAPLES	.088	.337	.566	.064	.264
U-Net	bstd	FGADR	FGADR	.084	.568	.634	.325	.403
U-Net	bstd	FGADR	Retinal-L.	.084	.224	.368	.155	.208
U-Net	bstd	FGADR	DiaRetDB1	.005	.261	.563	.188	.254
U-Net	bstd	TJDR-std	IDRiD	.108	.324	.641	.435	.377
U-Net	bstd	TJDR-std	DDR	.071	.264	.531	.161	.257
U-Net	bstd	TJDR-std	TJDR std	.326	.775	.642	.455	.550
U-Net	bstd	TJDR-std	TJDR UWF	.015	.203	.077	.182	.119
U-Net	bstd	TJDR-std	MAPLES	.080	.327	.447	.033	.222
U-Net	bstd	TJDR-std	FGADR	.015	.237	.266	.076	.148
U-Net	bstd	TJDR-std	Retinal-L.	.028	.107	.193	.098	.107
U-Net	bstd	TJDR-std	DiaRetDB1	.008	.169	.217	.144	.134
nnU-Net	bstd	FGADR	IDRiD	.121	.596	.826	.717	.565
nnU-Net	bstd	FGADR	DDR	.063	.378	.532	.306	.320
nnU-Net	bstd	FGADR	TJDR std	.147	.737	.696	.520	.525
nnU-Net	bstd	FGADR	TJDR UWF	.021	.277	.222	.391	.228
nnU-Net	bstd	FGADR	MAPLES	.197	.517	.719	.138	.393
nnU-Net	bstd	FGADR	FGADR	.136	.631	.659	.372	.449
nnU-Net	bstd	FGADR	Retinal-L.	.039	.390	.439	.247	.279
nnU-Net	bstd	FGADR	DiaRetDB1	.028	.474	.559	.226	.322
nnU-Net	bstd	TJDR-std	IDRiD	.146	.459	.664	.474	.436
nnU-Net	bstd	TJDR-std	DDR	.079	.310	.550	.170	.277
nnU-Net	bstd	TJDR-std	TJDR std	.305	.785	.787	.668	.636
nnU-Net	bstd	TJDR-std	TJDR UWF	.013	.311	.124	.268	.179
nnU-Net	bstd	TJDR-std	MAPLES	.162	.462	.457	.033	.278
nnU-Net	bstd	TJDR-std	FGADR	.019	.322	.299	.157	.199
nnU-Net	bstd	TJDR-std	Retinal-L.	.031	.162	.224	.125	.136
nnU-Net	bstd	TJDR-std	DiaRetDB1	.027	.395	.338	.157	.229
Annotation granularity								
M2MRF	DDR	DDR	IDRiD	.392	.555	.776	.563	.571
M2MRF	DDR	DDR	TJDR std	.299	.696	.618	.412	.506
M2MRF	DDR	DDR	DiaRetDB1	.049	.218	.177	.247	.173
U-Net	DDR	DDR	IDRiD	.364	.541	.800	.483	.547
U-Net	DDR	DDR	TJDR std	.296	.676	.614	.392	.494
U-Net	DDR	DDR	DiaRetDB1	.039	.314	.268	.253	.219
M2MRF	FGADR	FGADR	IDRiD	.141	.320	.729	.557	.437
M2MRF	FGADR	FGADR	TJDR std	.267	.642	.666	.223	.449
M2MRF	FGADR	FGADR	DiaRetDB1	.006	.270	.287	.142	.176
U-Net	FGADR	FGADR	IDRiD	.092	.294	.793	.276	.364
U-Net	FGADR	FGADR	TJDR std	.135	.718	.603	.339	.449
U-Net	FGADR	FGADR	DiaRetDB1	.005	.261	.563	.188	.254
M2MRF	DDR + FGADR (383)	DDR + FGADR (383)	IDRiD	.386	.446	.799	.594	.556
M2MRF	DDR + FGADR (383)	DDR + FGADR (383)	DDR	.141	.365	.508	.328	.336
M2MRF	DDR + FGADR (383)	DDR + FGADR (383)	TJDR std	.285	.669	.661	.321	.484
M2MRF	DDR + FGADR (383)	DDR + FGADR (383)	TJDR UWF	.110	.461	.172	.180	.231
M2MRF	DDR + FGADR (383)	DDR + FGADR (383)	MAPLES	.252	.378	.482	.134	.311
M2MRF	DDR + FGADR (383)	DDR + FGADR (383)	Retinal-L.	.044	.158	.177	.176	.139

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Model	Level	Trained on	Test set	MA	HE	EX	SE	mAUPR
M2MRF	DDR + FGADR (383)	DDR + FGADR (383)	DiaRetDB1	.043	.197	.222	.182	.161
M2MRF	DDR + FGADR	DDR + FGADR	IDRiD	.403	.466	.782	.565	.554
M2MRF	DDR + FGADR	DDR + FGADR	DDR	.139	.366	.492	.282	.320
M2MRF	DDR + FGADR	DDR + FGADR	TJDR std	.312	.680	.659	.337	.497
M2MRF	DDR + FGADR	DDR + FGADR	TJDR UWF	.109	.474	.186	.264	.258
M2MRF	DDR + FGADR	DDR + FGADR	MAPLES	.256	.391	.431	.119	.299
M2MRF	DDR + FGADR	DDR + FGADR	Retinal-L.	.045	.168	.173	.160	.137
M2MRF	DDR + FGADR	DDR + FGADR	DiaRetDB1	.041	.208	.176	.151	.144
M2MRF	DDR + Retinal-Lesions	DDR + Retinal-Lesions	IDRiD	.364	.426	.795	.533	.530
M2MRF	DDR + Retinal-Lesions	DDR + Retinal-Lesions	DDR	.120	.308	.479	.316	.306
M2MRF	DDR + Retinal-Lesions	DDR + Retinal-Lesions	TJDR std	.120	.539	.334	.276	.317
M2MRF	DDR + Retinal-Lesions	DDR + Retinal-Lesions	TJDR UWF	.078	.456	.208	.103	.211
M2MRF	DDR + Retinal-Lesions	DDR + Retinal-Lesions	MAPLES	.134	.295	.367	.137	.233
M2MRF	DDR + Retinal-Lesions	DDR + Retinal-Lesions	FGADR	.009	.170	.177	.153	.127
M2MRF	DDR + Retinal-Lesions	DDR + Retinal-Lesions	DiaRetDB1	.048	.231	.347	.243	.217
M2MRF	Retinal-Lesions	Retinal-Lesions	IDRiD	.063	.133	.186	.285	.167
M2MRF	Retinal-Lesions	Retinal-Lesions	DDR	.024	.097	.130	.166	.104
M2MRF	Retinal-Lesions	Retinal-Lesions	TJDR std	.039	.112	.160	.287	.150
M2MRF	Retinal-Lesions	Retinal-Lesions	TJDR UWF	.015	.121	.053	.120	.077
M2MRF	Retinal-Lesions	Retinal-Lesions	MAPLES	.110	.101	.342	.074	.157
M2MRF	Retinal-Lesions	Retinal-Lesions	FGADR	.018	.121	.131	.129	.100
M2MRF	Retinal-Lesions	Retinal-Lesions	Retinal-L.	.352	.508	.650	.366	.469
M2MRF	Retinal-Lesions	Retinal-Lesions	DiaRetDB1	.022	.328	.774	.355	.370
U-Net	DDR + FGADR (383)	DDR + FGADR (383)	IDRiD	.375	.517	.828	.594	.578
U-Net	DDR + FGADR (383)	DDR + FGADR (383)	DDR	.137	.414	.548	.376	.369
U-Net	DDR + FGADR (383)	DDR + FGADR (383)	TJDR std	.299	.728	.556	.177	.440
U-Net	DDR + FGADR (383)	DDR + FGADR (383)	TJDR UWF	.101	.473	.151	.242	.242
U-Net	DDR + FGADR (383)	DDR + FGADR (383)	MAPLES	.305	.472	.502	.057	.334
U-Net	DDR + FGADR (383)	DDR + FGADR (383)	Retinal-L.	.068	.244	.258	.177	.187
U-Net	DDR + FGADR (383)	DDR + FGADR (383)	DiaRetDB1	.045	.329	.352	.161	.221
U-Net	DDR + FGADR	DDR + FGADR	IDRiD	.365	.551	.858	.613	.597
U-Net	DDR + FGADR	DDR + FGADR	DDR	.117	.430	.545	.402	.373
U-Net	DDR + FGADR	DDR + FGADR	TJDR std	.325	.762	.622	.342	.513
U-Net	DDR + FGADR	DDR + FGADR	TJDR UWF	.089	.466	.210	.294	.265
U-Net	DDR + FGADR	DDR + FGADR	MAPLES	.257	.468	.575	.133	.358
U-Net	DDR + FGADR	DDR + FGADR	Retinal-L.	.042	.219	.236	.186	.171
U-Net	DDR + FGADR	DDR + FGADR	DiaRetDB1	.034	.320	.302	.173	.207
U-Net	DDR + Retinal-Lesions	DDR + Retinal-Lesions	IDRiD	.279	.502	.786	.521	.522
U-Net	DDR + Retinal-Lesions	DDR + Retinal-Lesions	DDR	.090	.366	.518	.389	.341
U-Net	DDR + Retinal-Lesions	DDR + Retinal-Lesions	TJDR std	.237	.643	.490	.440	.452
U-Net	DDR + Retinal-Lesions	DDR + Retinal-Lesions	TJDR UWF	.057	.480	.231	.210	.245
U-Net	DDR + Retinal-Lesions	DDR + Retinal-Lesions	MAPLES	.269	.465	.474	.147	.339
U-Net	DDR + Retinal-Lesions	DDR + Retinal-Lesions	FGADR	.013	.202	.271	.133	.155
U-Net	DDR + Retinal-Lesions	DDR + Retinal-Lesions	DiaRetDB1	.052	.278	.386	.278	.248
U-Net	Retinal-Lesions	Retinal-Lesions	IDRiD	.091	.230	.353	.317	.248
U-Net	Retinal-Lesions	Retinal-Lesions	DDR	.039	.146	.239	.234	.164
U-Net	Retinal-Lesions	Retinal-Lesions	TJDR std	.087	.378	.238	.419	.281
U-Net	Retinal-Lesions	Retinal-Lesions	TJDR UWF	.026	.208	.091	.163	.122
U-Net	Retinal-Lesions	Retinal-Lesions	MAPLES	.168	.284	.470	.102	.256
U-Net	Retinal-Lesions	Retinal-Lesions	FGADR	.017	.136	.148	.064	.091
U-Net	Retinal-Lesions	Retinal-Lesions	Retinal-L.	.353	.553	.721	.390	.504
U-Net	Retinal-Lesions	Retinal-Lesions	DiaRetDB1	.045	.494	.874	.483	.474

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Model	Level	Trained on	Test set	MA	HE	EX	SE	mAUPR
Input resolution								
M2MRF	res:1440	DDR	DDR	.148	.402	.547	.422	.380
RETFound + UPerNet	res:1440	DDR	DDR	.189	.491	.640	.546	.466
HRDecoder	res:2880 (author canvas, equivalent epochs)	IDRiD	IDRiD	.444	.596	.853	.676	.642
M2MRF	res:512	DDR	DDR	.111	.371	.548	.382	.353
M2MRF	res:native patches	DDR	DDR	.140	.360	.510	.237	.312
RETFound + UPerNet	res:512	DDR	DDR	.083	.500	.614	.568	.441
RETFound + UPerNet	res:native patches	DDR	DDR	.197	.502	.620	.471	.448
Training amount								
M2MRF	size:DDR 100%	DDR	DDR	.148	.402	.547	.422	.380
M2MRF	size:DDR 100%	DDR	TJDR std	.299	.696	.618	.412	.506
M2MRF	size:DDR 100%	DDR	FGADR	.019	.310	.340	.181	.212
U-Net	size:DDR 100%	DDR	DDR	.141	.410	.553	.410	.378
U-Net	size:DDR 100%	DDR	TJDR std	.296	.676	.614	.392	.494
U-Net	size:DDR 100%	DDR	FGADR	.013	.214	.292	.162	.170
M2MRF	size:DDR 10%	DDR	IDRiD	.199	.432	.741	.409	.445
M2MRF	size:DDR 10%	DDR	DDR	.097	.356	.521	.209	.296
M2MRF	size:DDR 10%	DDR	TJDR std	.220	.600	.590	.281	.423
M2MRF	size:DDR 10%	DDR	TJDR UWF	.103	.248	.155	.035	.135
M2MRF	size:DDR 10%	DDR	MAPLES	.192	.318	.437	.050	.249
M2MRF	size:DDR 10%	DDR	FGADR	.010	.209	.178	.143	.135
M2MRF	size:DDR 10%	DDR	Retinal-L.	.037	.114	.172	.136	.115
M2MRF	size:DDR 10%	DDR	DiaRetDB1	.026	.159	.194	.195	.144
M2MRF	size:DDR 50%	DDR	IDRiD	.370	.456	.779	.571	.544
M2MRF	size:DDR 50%	DDR	DDR	.151	.341	.523	.244	.315
M2MRF	size:DDR 50%	DDR	TJDR std	.286	.690	.572	.309	.464
M2MRF	size:DDR 50%	DDR	TJDR UWF	.105	.364	.196	.052	.179
M2MRF	size:DDR 50%	DDR	MAPLES	.240	.425	.465	.112	.311
M2MRF	size:DDR 50%	DDR	FGADR	.017	.274	.315	.212	.204
M2MRF	size:DDR 50%	DDR	Retinal-L.	.043	.189	.179	.155	.141
M2MRF	size:DDR 50%	DDR	DiaRetDB1	.045	.237	.179	.216	.169
M2MRF	size:fine merged	DDR	IDRiD	.446	.531	.824	.674	.619
M2MRF	size:fine merged	DDR	DDR	.141	.393	.534	.405	.368
M2MRF	size:fine merged	DDR	TJDR std	.333	.746	.640	.540	.565
M2MRF	size:fine merged	DDR	TJDR UWF	.120	.472	.197	.339	.282
M2MRF	size:fine merged	DDR	MAPLES	.285	.462	.500	.150	.349
M2MRF	size:fine merged	DDR	FGADR	.020	.347	.333	.194	.224
M2MRF	size:fine merged	DDR	Retinal-L.	.049	.191	.184	.187	.153
M2MRF	size:fine merged	DDR	DiaRetDB1	.049	.266	.191	.256	.190
U-Net	size:DDR 10%	DDR	IDRiD	.239	.474	.784	.470	.492
U-Net	size:DDR 10%	DDR	DDR	.106	.337	.508	.334	.321
U-Net	size:DDR 10%	DDR	TJDR std	.197	.525	.479	.308	.378
U-Net	size:DDR 10%	DDR	TJDR UWF	.077	.243	.205	.072	.149
U-Net	size:DDR 10%	DDR	MAPLES	.243	.303	.542	.092	.295
U-Net	size:DDR 10%	DDR	FGADR	.010	.132	.190	.119	.112
U-Net	size:DDR 10%	DDR	Retinal-L.	.037	.119	.173	.165	.123
U-Net	size:DDR 10%	DDR	DiaRetDB1	.030	.180	.247	.204	.165
U-Net	size:DDR 50%	DDR	IDRiD	.303	.552	.811	.493	.540
U-Net	size:DDR 50%	DDR	DDR	.132	.407	.570	.383	.373
U-Net	size:DDR 50%	DDR	TJDR std	.273	.715	.530	.374	.473
U-Net	size:DDR 50%	DDR	TJDR UWF	.123	.338	.261	.075	.199
U-Net	size:DDR 50%	DDR	MAPLES	.240	.476	.564	.076	.339

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Model	Level	Trained on	Test set	MA	HE	EX	SE	mAUPR
U-Net	size:DDR 50%	DDR	FGADR	.008	.230	.355	.155	.187
U-Net	size:DDR 50%	DDR	Retinal-L.	.045	.192	.214	.189	.160
U-Net	size:DDR 50%	DDR	DiaRetDB1	.035	.307	.262	.234	.210
U-Net	size:fine merged	DDR	IDRiD	.432	.563	.859	.574	.607
U-Net	size:fine merged	DDR	DDR	.154	.393	.582	.337	.367
U-Net	size:fine merged	DDR	TJDR std	.367	.779	.643	.458	.562
U-Net	size:fine merged	DDR	TJDR UWF	.118	.489	.179	.288	.268
U-Net	size:fine merged	DDR	MAPLES	.295	.491	.596	.107	.372
U-Net	size:fine merged	DDR	FGADR	.014	.276	.322	.154	.192
U-Net	size:fine merged	DDR	Retinal-L.	.050	.229	.240	.191	.177
U-Net	size:fine merged	DDR	DiaRetDB1	.041	.338	.292	.202	.218
M2MRF	size:DDR 10%:fov	DDR	TJDR UWF	.064	.235	.100	.051	.113
M2MRF	size:DDR 50%:fov	DDR	TJDR UWF	.115	.437	.151	.165	.217
U-Net	size:DDR 10%:fov	DDR	TJDR UWF	.047	.238	.080	.121	.122
U-Net	size:DDR 50%:fov	DDR	TJDR UWF	.065	.412	.187	.221	.221
Loss								
U-Net	loss:CE + Dice	DDR	DDR	.141	.410	.553	.410	.378
U-Net	loss:CE	DDR	DDR	.126	.429	.602	.271	.357
U-Net	loss:Dice	DDR	DDR	.124	.268	.397	.160	.237
U-Net	loss:CE + per-image Dice	DDR	DDR	.006	.010	.460	.000	.119

Supplementary Table 8 Computational cost. Parameters and trainable parameters under the unified protocol, training time of one seed on each source (40 GB A100, one task per card), and the inference cost of one 1,440-pixel canvas (a 1440×960 field of view) with the model's own inference geometry: floating-point operations (convolutions and matrix products, all windows), peak GPU memory allocated during inference, and the median wall time of five passes on an A100.

Model	Params (M)	Trainable (M)	GPU h IDRiD	GPU h DDR	GFLOPs	Peak infer. mem. (GB)	Infer. (s/image)
U-Net	7.8	7.9	1.5	5.4	589	1.4	.03
DeepLabv3	58.6	58.6	3.6	12.8	2,543	.9	.05
HRNet-W48	65.8	65.9	2.6	10.1	984	1.3	.05
Swin-Unet	27.2	27.2	2.6	10.4	1,169	.5	.17
FCT	1.9	1.9	19.7	72.0	2,942	.3	.94
nnU-Net	59.2	–	7.2	7.2	–	–	–
nnU-Net ResEnc-M	105.9	–	12.8	11.4	–	–	–
M2MRF	70.3	70.3	3.0	11.6	931	1.1	.05
HACDR-Net	24.6	–	5.9	27.8	–	–	–
HRDecoder	65.6	–	1.5	5.9	–	–	–
PMCNet	12.9	12.9	1.8	7.1	443	1.9	.03
WSRFNet	9.7	9.7	2.7	10.5	831	1.4	.05
CLC-Net	119.7	119.7	9.9	39.2	11,760	3.9	.86
TC-Net	14.9	14.9	6.0	24.7	9,631	8.7	.29
SAMed	91.6	4.4	2.1	7.4	–	1.9	.30
SAM-Adapter	641.3	4.2	6.5	26.3	11,960	7.4	.92
SAM2-UNet	216.5	4.4	4.1	16.2	2,677	1.6	.21
MedSAM (full FT)	93.7	93.7	2.8	11.1	1,951	5.5	.19
RETFound + UPerNet	352.6	352.6	6.1	24.2	20,870	2.5	1.10

nnU-Net follows its own schedule of 1,000 epochs rather than the pixel budget and runs its own inference pipeline, so it has no audited inference columns. HACDR-Net and HRDecoder could not be built in the audit environment (their mmcv forks need another environment) and were not measured. SAMed's FLOPs were not counted (its forward is not traced by the counter). Parameters of nnU-Net are those of its plans.

Supplementary Table 9 Refined IDRiD (label revision): per-class AUPR of every IDRiD-trained model against the official and the refined labels through the same 1,024-pixel geometry. Test maps of the main-table checkpoints (seed 0), moved through the recovered crop and scored with the protocol evaluator against both label sets; no model is retrained or re-run.

Model	MA off.	MA ref.	HE off.	HE ref.	EX off.	EX ref.	SE off.	SE ref.	mAUPR off.	mAUPR ref.	Δ
CLC-Net	.423	.420	.686	.685	.890	.889	.641	.640	.660	.659	−0.001
DeepLabv3	.091	.091	.643	.643	.736	.736	.664	.664	.533	.533	+0.000
FCT	.079	.078	.165	.165	.687	.686	.115	.115	.261	.261	−0.000
HACDR-Net	.358	.356	.504	.504	.828	.827	.561	.560	.563	.561	−0.001
HRDecoder	.407	.405	.634	.633	.827	.826	.730	.730	.649	.649	−0.001
HRNet-W48	.442	.438	.685	.684	.885	.883	.783	.783	.699	.697	−0.002
M2MRF	.439	.436	.625	.625	.879	.878	.681	.680	.656	.655	−0.001
MedSAM (full FT)	.370	.368	.666	.666	.858	.858	.676	.676	.643	.642	−0.001
PMCNet	.411	.409	.610	.610	.883	.881	.698	.698	.651	.649	−0.001
RETFound + UPerNet	.483	.480	.719	.718	.896	.895	.736	.736	.709	.707	−0.001
SAM2-UNet	.423	.421	.683	.682	.875	.874	.681	.681	.666	.665	−0.001
SAM-Adapter	.442	.439	.745	.745	.887	.887	.756	.756	.708	.707	−0.001
SAMed	.445	.442	.737	.737	.893	.892	.791	.790	.717	.715	−0.001
Swin-Unet	.473	.471	.712	.712	.895	.894	.689	.689	.692	.691	−0.001
TC-Net	.353	.351	.651	.651	.855	.854	.755	.755	.654	.653	−0.001
U-Net	.386	.384	.623	.622	.895	.894	.710	.711	.653	.653	−0.001
WSRFNet	.388	.386	.602	.602	.856	.855	.506	.506	.588	.587	−0.001
nnU-Net	.482	.480	.673	.673	.878	.876	.645	.644	.669	.668	−0.001
nnU-Net ResEnc-M	.442	.440	.667	.666	.878	.877	.541	.540	.632	.631	−0.001

Label differences over the 27 test images: MA: 13 images changed, 28,847 → 28,886 labelled pixels, 1,083 → 1,083 components; HE: 13 images changed, 311,289 → 311,524 labelled pixels, 532 → 532 components; EX: 16 images changed, 320,363 → 320,160 labelled pixels, 3,895 → 3,882 components; SE: 6 images changed, 53,684 → 53,597 labelled pixels, 38 → 38 components. The crop geometry of every image (all 27: padded to a square and resized; NCC $\geq$ 0.9988) is runs/REFINED/RefinedIDRiD_geometry.json.

Supplementary Table 10 Encoder comparison: four ViT-L encoders under one UPerNet head, recipe and pixel budget (seed 0). Per-class AUPR on the internal test and on the three independent fine-grained sets. RETFound-MAE is the main-table RETFound run of the same source; the other three are the control cells. The pre-specified contrasts are RETFound-MAE minus MAE-IN1k and RETFound-DINOv2 minus DINOv2.

Encoder	Test set	MA	HE	EX	SE	mAUPR
Trained on IDRiD						
RETFound-DINOv2	IDRiD	.502	.759	.900	.683	.711
RETFound-DINOv2	TJDR std	.375	.626	.714	.520	.559
RETFound-DINOv2	TJDR UWF	.092	.484	.211	.104	.223
RETFound-DINOv2	MAPLES	.380	.495	.682	.109	.417
RETFound-MAE	IDRiD	.484	.718	.895	.734	.708
RETFound-MAE	TJDR std	.393	.582	.776	.070	.455
RETFound-MAE	TJDR UWF	.099	.494	.106	.311	.253
RETFound-MAE	MAPLES	.375	.503	.680	.092	.413
DINOv2 ViT-L	IDRiD	.515	.765	.905	.772	.739
DINOv2 ViT-L	TJDR std	.341	.658	.676	.427	.525
DINOv2 ViT-L	TJDR UWF	.102	.557	.239	.091	.247
DINOv2 ViT-L	MAPLES	.392	.490	.693	.194	.442
MAE-IN1k ViT-L	IDRiD	.493	.744	.900	.778	.729
MAE-IN1k ViT-L	TJDR std	.392	.547	.709	.507	.539
MAE-IN1k ViT-L	TJDR UWF	.099	.586	.244	.147	.269
MAE-IN1k ViT-L	MAPLES	.370	.469	.659	.203	.425
Trained on DDR						
RETFound-DINOv2	DDR	.184	.508	.657	.611	.490
RETFound-DINOv2	TJDR std	.354	.786	.716	.607	.616
RETFound-DINOv2	TJDR UWF	.113	.264	.187	.042	.151
RETFound-DINOv2	MAPLES	.422	.522	.731	.307	.495
RETFound-MAE	DDR	.189	.491	.640	.546	.466
RETFound-MAE	TJDR std	.387	.733	.724	.513	.589
RETFound-MAE	TJDR UWF	.073	.064	.278	.030	.111
RETFound-MAE	MAPLES	.420	.531	.722	.166	.460
DINOv2 ViT-L	DDR	.183	.572	.668	.618	.510
DINOv2 ViT-L	TJDR std	.390	.791	.698	.596	.619
DINOv2 ViT-L	TJDR UWF	.100	.133	.265	.050	.137
DINOv2 ViT-L	MAPLES	.425	.523	.728	.274	.488
MAE-IN1k ViT-L	DDR	.206	.567	.670	.634	.519
MAE-IN1k ViT-L	TJDR std	.422	.788	.717	.611	.634
MAE-IN1k ViT-L	TJDR UWF	.103	.100	.244	.061	.127
MAE-IN1k ViT-L	MAPLES	.406	.515	.726	.265	.478

Supplementary Table 11 Ensembles of the main-table probability maps and the reference model, mAUPR on every test set. Compositions fixed before any ensemble was scored (Methods): all 18 registered members, the five and three best by validation mAUPR, the best of each family, and each family alone. The reference model's two cells are trained on the three fine-grained sources merged; their ensemble is the pixel mean of their maps.

Composition	IDRiD	DDR	TJDR std	TJDR UWF	MAPLES	FGADR	Retinal-L.	DiaRetDB1
Members trained on IDRiD								
E-all	.764	.400	.594	.311	.455	.279[P]	.216[P]	.317
E-top5	.754	.410	.598	.292	.449	.290	.251	.316
E-top3	.742	.394	.559	.279	.437	.273	.252	.325
E-1perfam	.741	.381	.555	.284	.445	.256	.239	.318
E-fam:general	.727	.352	.513	.276	.410	.244	.194	.302
E-fam:auto	.693	.316	.485	.229	.406	.242	.177	.269
E-fam:fundus	.736	.371	.568	.292	.436	.250[P]	.214[P]	.300
E-fam:foundation	.757	.411	.609	.297	.442	.294	.250	.318
Members trained on DDR								
E-all	.677	.505	.637	.266	.496	.291[P]	.278[P]	.335
E-top5	.680	.511	.652	.242	.506	.298	.290	.335
E-top3	.672	.510	.655	.226	.497	.300	.291	.339
E-1perfam	.665	.494	.625	.228	.486	.281	.278	.321
E-fam:general	.650	.483	.606	.227	.470	.268	.266	.324
E-fam:auto	.641	.463	.572	.210	.473	.275	.227	.334
E-fam:fundus	.637	.444	.568	.246	.452	.251[P]	.218[P]	.305
E-fam:foundation	.676	.509	.657	.248	.494	.302	.294	.340
Reference model: trained on IDRiD + DDR + TJDR std merged (DDR budget and validation)								
RETFound + UPerNet (fine merged)	.672	.468	.704	.190	.463	.272	.282	.343
SAMed (fine merged)	.674	.470	.638	.314	.458	.306	.260	.318
RetiSeg-Ref (ensemble)	.710	.501	.696	.302	.482	.309	.278	.339

[P], computed without HACDR-Net, whose maps for that set could not be regenerated (members named in the run record). In-source test sets of the reference model (IDRiD, DDR, TJDR std) are the official test splits of sources that entered its training.

Supplementary Table 12 Results by DR grade (seed 0). On the test sets that carry a DR grade per image, mAUPR and image-level AUC recomputed within grades 0–1 and within grades 2–4 from the stored per-image histograms, with the thresholds of the full test; ungradable images enter neither stratum.

Model	Trained on	mAUPR 0–1	mAUPR 2–4	Image AUC 0–1	Image AUC 2–4	Δ mAUPR
DDR: 20 images of grades 0–1, 203 of grades 2–4						
CLC-Net	DDR	.254	.384	.653	.752	+0.129
CLC-Net	IDRiD	.147	.295	.486	.667	+0.148
DeepLabv3	DDR	.145	.372	–	–	+0.227
DeepLabv3	IDRiD	.169	.258	.486	.720	+0.090
FCT	DDR	.001	.177	–	–	+0.176
FCT	IDRiD	.002	.078	.486	.597	+0.076
HACDR-Net	DDR	.210	.268	–	–	+0.058
HACDR-Net	IDRiD	.196	.163	.514	.651	−0.032
HRDecoder	DDR	.266	.273	.639	.692	+0.007
HRDecoder	IDRiD	.128	.276	.542	.658	+0.148
HRNet-W48	DDR	.260	.442	–	–	+0.182
HRNet-W48	IDRiD	.257	.336	.444	.772	+0.079
M2MRF	DDR	.209	.349	–	–	+0.140
M2MRF	IDRiD	.120	.218	.542	.628	+0.099
MedSAM (full FT)	DDR	.225	.361	.444	.782	+0.137
MedSAM (full FT)	IDRiD	.189	.315	.639	.757	+0.126
PMCNet	DDR	.213	.302	.611	.750	+0.089
PMCNet	IDRiD	.193	.262	.167	.701	+0.069
RETFound + UPerNet	DDR	.256	.462	.597	.829	+0.206
RETFound + UPerNet	IDRiD	.183	.355	.292	.797	+0.172
SAM2-UNet	DDR	.270	.315	.514	.741	+0.044
SAM2-UNet	IDRiD	.156	.283	.604	.615	+0.127
SAM-Adapter	DDR	.273	.416	.583	.639	+0.143
SAM-Adapter	IDRiD	.189	.292	.625	.590	+0.103
SAMed	DDR	.227	.466	.361	.832	+0.240
SAMed	IDRiD	.189	.381	.576	.778	+0.192
Swin-Unet	DDR	.251	.431	–	–	+0.179
Swin-Unet	IDRiD	.157	.329	.340	.744	+0.172
TC-Net	DDR	.185	.360	.417	.761	+0.176
TC-Net	IDRiD	.197	.313	.500	.772	+0.116
U-Net	DDR	.206	.341	.458	.776	+0.135
U-Net	IDRiD	.112	.265	.222	.715	+0.153
WSRFNet	DDR	.160	.217	.569	.637	+0.057
WSRFNet	IDRiD	.123	.236	.479	.609	+0.113
nnU-Net	DDR	.150	.410	–	–	+0.260
nnU-Net	IDRiD	.124	.274	.514	.702	+0.150
nnU-Net ResEnc-M	DDR	.151	.403	–	–	+0.252
nnU-Net ResEnc-M	IDRiD	.150	.249	.542	.750	+0.098
MAPLES: 56 images of grades 0–1, 4 of grades 2–4						
CLC-Net	DDR	.297	.308	.650	.667	+0.011
CLC-Net	IDRiD	.349	.445	.638	1.000	+0.096
DeepLabv3	DDR	.332	.366	.752	1.000	+0.034
DeepLabv3	IDRiD	.261	.425	.693	1.000	+0.164
FCT	DDR	.127	.186	.470	.333	+0.060
FCT	IDRiD	.107	.177	.521	1.000	+0.069
HACDR-Net	DDR	.290	.371	.613	1.000	+0.081
HACDR-Net	IDRiD	.265	.407	.635	1.000	+0.143
HRDecoder	DDR	.265	.284	.761	.833	+0.019
HRDecoder	IDRiD	.338	.436	.650	1.000	+0.098
HRNet-W48	DDR	.433	.459	.788	.667	+0.027
HRNet-W48	IDRiD	.379	.465	.638	1.000	+0.086
M2MRF	DDR	.292	.329	.690	1.000	+0.037
M2MRF	IDRiD	.280	.330	.638	1.000	+0.050
MedSAM (full FT)	DDR	.373	.425	.743	.667	+0.051
MedSAM (full FT)	IDRiD	.337	.478	.738	1.000	+0.142
PMCNet	DDR	.370	.434	.735	.667	+0.064
PMCNet	IDRiD	.325	.436	.679	1.000	+0.110
RETFound + UPerNet	DDR	.437	.494	.830	.667	+0.056
RETFound + UPerNet	IDRiD	.386	.509	.721	.667	+0.123
SAM2-UNet	DDR	.354	.403	.675	.833	+0.049
SAM2-UNet	IDRiD	.332	.391	.658	.833	+0.059
SAM-Adapter	DDR	.325	.358	.657	.833	+0.032
SAM-Adapter	IDRiD	.334	.350	.677	1.000	+0.016
SAMed	DDR	.454	.471	.807	1.000	+0.018
SAMed	IDRiD	.393	.495	.792	1.000	+0.101

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Model	Trained on	mAUPR 0–1	mAUPR 2–4	Image AUC 0–1	Image AUC 2–4	Δ mAUPR
Swin-Unet	DDR	.435	.548	.770	1.000	+0.113
Swin-Unet	IDRiD	.374	.456	.622	.667	+0.083
TC-Net	DDR	.324	.432	.735	.667	+0.108
TC-Net	IDRiD	.383	.522	.701	1.000	+0.138
U-Net	DDR	.262	.375	.699	1.000	+0.114
U-Net	IDRiD	.277	.374	.681	1.000	+0.097
WSRFNet	DDR	.257	.327	.669	.667	+0.071
WSRFNet	IDRiD	.264	.319	.630	.333	+0.054
nnU-Net	DDR	.343	.427	.818	1.000	+0.084
nnU-Net	IDRiD	.338	.420	.700	1.000	+0.082
nnU-Net ResEnc-M	DDR	.375	.423	.797	1.000	+0.048
nnU-Net ResEnc-M	IDRiD	.301	.350	.696	.333	+0.050
FGADR: 94 images of grades 0–1, 459 of grades 2–4						
CLC-Net	DDR	.059	.166	.560	.747	+0.107
CLC-Net	IDRiD	.016	.172	.633	.698	+0.156
DeepLabv3	DDR	.056	.236	.762	.753	+0.179
DeepLabv3	IDRiD	.055	.222	.661	.762	+0.167
FCT	DDR	.003	.071	.571	.602	+0.068
FCT	IDRiD	.002	.052	.577	.634	+0.050
HACDR-Net	DDR	.012	.106	.443	.674	+0.095
HACDR-Net	IDRiD	.005	.038	.501	.573	+0.032
HRDecoder	DDR	.021	.188	.707	.727	+0.168
HRDecoder	IDRiD	.024	.199	.650	.715	+0.176
HRNet-W48	DDR	.046	.274	.775	.814	+0.229
HRNet-W48	IDRiD	.036	.256	.759	.802	+0.220
M2MRF	DDR	.037	.202	.714	.731	+0.165
M2MRF	IDRiD	.002	.084	.581	.599	+0.082
MedSAM (full FT)	DDR	.074	.231	.768	.803	+0.157
MedSAM (full FT)	IDRiD	.031	.270	.635	.759	+0.238
PMCNet	DDR	.036	.211	.610	.793	+0.175
PMCNet	IDRiD	.027	.216	.730	.752	+0.189
RETFound + UPerNet	DDR	.061	.268	.700	.814	+0.207
RETFound + UPerNet	IDRiD	.017	.240	.719	.773	+0.222
SAM2-UNet	DDR	.028	.222	.712	.754	+0.195
SAM2-UNet	IDRiD	.037	.235	.734	.759	+0.198
SAM-Adapter	DDR	.035	.242	.687	.669	+0.207
SAM-Adapter	IDRiD	.014	.221	.564	.620	+0.207
SAMed	DDR	.067	.295	.770	.832	+0.228
SAMed	IDRiD	.053	.289	.745	.800	+0.236
Swin-Unet	DDR	.043	.248	.700	.802	+0.204
Swin-Unet	IDRiD	.017	.204	.590	.726	+0.187
TC-Net	DDR	.039	.219	.735	.795	+0.180
TC-Net	IDRiD	.020	.221	.640	.782	+0.201
U-Net	DDR	.019	.170	.690	.761	+0.151
U-Net	IDRiD	.003	.144	.576	.700	+0.141
WSRFNet	DDR	.016	.062	.652	.639	+0.046
WSRFNet	IDRiD	.016	.187	.642	.651	+0.170
nnU-Net	DDR	.046	.249	.728	.773	+0.203
nnU-Net	IDRiD	.036	.222	.673	.784	+0.186
nnU-Net ResEnc-M	DDR	.056	.245	.794	.810	+0.190
nnU-Net ResEnc-M	IDRiD	.027	.189	.615	.773	+0.162
Retinal-L.: 150 images of grades 0–1, 328 of grades 2–4						
CLC-Net	DDR	.057	.125	.762	.807	+0.068
CLC-Net	IDRiD	.029	.168	.807	.768	+0.139
DeepLabv3	DDR	.097	.273	.730	.798	+0.176
DeepLabv3	IDRiD	.051	.286	.695	.746	+0.236
FCT	DDR	.003	.085	.612	.605	+0.081
FCT	IDRiD	.002	.035	.529	.570	+0.033
HACDR-Net	DDR	.038	.140	.670	.782	+0.103
HACDR-Net	IDRiD	.012	.086	.684	.662	+0.074
HRDecoder	DDR	.050	.116	.849	.793	+0.066
HRDecoder	IDRiD	.030	.130	.767	.760	+0.100
HRNet-W48	DDR	.112	.264	.792	.846	+0.152
HRNet-W48	IDRiD	.061	.217	.780	.814	+0.156
M2MRF	DDR	.068	.135	.872	.831	+0.067
M2MRF	IDRiD	.023	.115	.744	.727	+0.092
MedSAM (full FT)	DDR	.058	.225	.834	.839	+0.167
MedSAM (full FT)	IDRiD	.042	.240	.790	.786	+0.197
PMCNet	DDR	.055	.179	.808	.833	+0.125
PMCNet	IDRiD	.039	.185	.740	.754	+0.146

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Model	Trained on	mAUPR 0–1	mAUPR 2–4	Image AUC 0–1	Image AUC 2–4	Δ mAUPR
RETFound + UPerNet	DDR	.121	.248	.826	.856	+0.127
RETFound + UPerNet	IDRiD	.040	.221	.774	.833	+0.182
SAM2-UNet	DDR	.081	.194	.807	.777	+0.113
SAM2-UNet	IDRiD	.034	.168	.754	.730	+0.134
SAM-Adapter	DDR	.074	.169	.789	.730	+0.094
SAM-Adapter	IDRiD	.020	.170	.612	.694	+0.151
SAMed	DDR	.125	.281	.828	.864	+0.155
SAMed	IDRiD	.055	.268	.795	.806	+0.213
Swin-Unet	DDR	.100	.263	.846	.870	+0.163
Swin-Unet	IDRiD	.052	.247	.810	.822	+0.195
TC-Net	DDR	.034	.182	.752	.808	+0.148
TC-Net	IDRiD	.026	.193	.768	.805	+0.167
U-Net	DDR	.069	.123	.850	.840	+0.054
U-Net	IDRiD	.028	.126	.778	.761	+0.098
WSRFNet	DDR	.032	.070	.705	.708	+0.037
WSRFNet	IDRiD	.036	.124	.756	.695	+0.087
nnU-Net	DDR	.073	.165	.858	.837	+0.092
nnU-Net	IDRiD	.028	.161	.835	.797	+0.132
nnU-Net ResEnc-M	DDR	.089	.172	.781	.873	+0.082
nnU-Net ResEnc-M	IDRiD	.020	.126	.798	.821	+0.106

Image-level AUC within a stratum is undefined when every image of the stratum carries the class or none does; such cells read 0 or 1 in the record and should be read with the stratum size.

Supplementary Table 13 Human reference on DiaRetDB1: each expert against the consensus of the other three. The expert's own markings, with their confidence levels mapped to scores, are the probability map and the consensus of the other three experts (≥ 0.75) is the ground truth; thresholds come from the official training list. A map with four confidence levels gives a four-point precision–recall curve, so the pixel-level AUPR is a lower bound.

Reader	MA	HE	EX	SE	mAUPR	Image AUC	Dice
Expert 01	.008	.246	.414	.102	.193	.870	.349
Expert 02	.001	.166	.273	.162	.150	.931	.261
Expert 03	.015	.341	.516	.244	.279	.931	.448
Expert 04	.022	.271	.366	.220	.219	.913	.360
Models trained on IDRiD, against the four-expert consensus (median over 19 models; range in brackets)							
models	–	–	–	–	0.245 [0.10–0.32]	0.833 [0.71–0.92]	–
Models trained on DDR, against the four-expert consensus (median over 19 models; range in brackets)							
models	–	–	–	–	0.284 [0.09–0.32]	0.840 [0.74–0.94]	–

Supplementary Table 14 Annotation style of every source: connected components per image and their size. All images of each source, every id once; 8-connected components of the prepared masks with ignore regions removed; the relative diameter is the component's equivalent diameter divided by the long side of the field-of-view box, in percent (Methods, annotation style).

Source	Class	Images	Components	Per image	Per positive image	Positive images	Diam. p10 (%)	p50 (%)	p90 (%)	Area px (median)
IDRiD	MA	81	3,497	43.2	43.2	81	0.33	0.50	0.78	234
IDRiD	HE	81	1,900	23.5	23.8	80	0.70	1.32	3.29	1608
IDRiD	EX	81	11,642	143.7	143.7	81	0.26	0.48	1.20	217
IDRiD	SE	81	150	1.9	3.8	40	1.86	3.07	5.45	8653
DDR	MA	757	10,380	13.7	18.2	570	0.21	0.43	0.77	48
DDR	HE	757	12,538	16.6	20.9	601	0.27	0.87	2.36	209
DDR	EX	757	23,659	31.3	48.7	486	0.16	0.45	1.31	56
DDR	SE	757	1,293	1.7	5.4	239	0.24	1.91	3.83	855
TJDR_std	MA	257	742	2.9	7.5	99	0.41	0.80	1.16	210
TJDR_std	HE	257	1,192	4.6	8.1	148	0.94	1.78	3.85	1048
TJDR_std	EX	257	1,130	4.4	11.0	103	0.62	1.26	2.76	525
TJDR_std	SE	257	216	0.8	3.3	65	1.36	3.11	6.19	3178
TJDR_uwf	MA	304	765	2.5	10.2	75	0.21	0.35	0.59	148
TJDR_uwf	HE	304	2,012	6.6	11.8	170	0.35	0.84	2.14	841
TJDR_uwf	EX	304	2,222	7.3	10.1	219	0.15	0.40	1.19	188
TJDR_uwf	SE	304	287	0.9	2.3	127	0.28	1.53	2.64	2826
FGADR	MA	1,842	11,235	6.1	7.9	1424	0.46	0.78	3.05	79
FGADR	HE	1,842	37,958	20.6	26.1	1456	0.59	1.17	3.45	176
FGADR	EX	1,842	52,896	28.7	41.4	1279	0.37	0.76	1.77	75
FGADR	SE	1,842	2,087	1.1	3.3	627	1.09	3.21	6.40	1323
Retinal-Lesions	MA	1,593	14,873	9.3	11.0	1346	1.37	2.31	4.10	336
Retinal-Lesions	HE	1,593	13,305	8.4	12.5	1064	1.82	3.45	7.12	749
Retinal-Lesions	EX	1,593	3,416	2.1	4.2	807	1.87	4.16	12.49	1089
Retinal-Lesions	SE	1,593	1,267	0.8	2.7	471	2.94	5.08	8.81	1624
DiaRetDB1	MA	89	183	2.1	4.6	40	0.48	0.62	0.72	61
DiaRetDB1	HE	89	402	4.5	10.1	40	0.71	2.93	8.61	1363
DiaRetDB1	EX	89	137	1.5	3.6	38	0.71	4.06	16.01	2624
DiaRetDB1	SE	89	27	0.3	1.8	15	1.90	4.31	8.02	2945
MAPLES-DR	MA	198	5,412	27.3	28.3	191	0.63	0.84	1.29	74
MAPLES-DR	HE	198	946	4.8	7.8	122	1.06	1.89	3.19	446
MAPLES-DR	EX	198	3,080	15.6	29.6	104	0.65	0.99	2.07	122
MAPLES-DR	SE	198	199	1.0	3.3	61	1.29	2.48	4.29	620

Supplementary Table 15 Near-duplicate audit between training sources and test sets, and the images excluded. Vessel signatures (green channel of the field-of-view box at 192 × 192, 5 × 5 black top-hat, normalised cross-correlation inside the field of view, the larger of the direct and the mirrored comparison) are correlated between every train or validation image of a source and every test image; a test image at or above the pair's threshold is left out of that source's independent tests (default 0.5; 0.33 within FGADR, whose images are near-duplicates of each other at scale). The pretraining corpora of the foundation models are not searched image by image; their overlap with the test sets is determined from the sources (Methods).

Training source	Test set	Excluded images
TJDR_uwf	TJDR_uwf	2
FGADR	FGADR	106
FGADR	Retinal-Lesions	19
Retinal-Lesions	FGADR	27
Retinal-Lesions	Retinal-Lesions	1
IDRiD	every test set	0
DDR	every test set	0
TJDR_std	every test set	0

Audit: bench/scripts/leak_audit.py –min-vessel 0.3 (all seven prepared sources), 4923 images. Strongest pairs (vessel NCC): FGADR train:0224_1 ↔ FGADR 0271_1 (1.00); FGADR train:0322_1 ↔ FGADR 0741_1 (1.00); FGADR train:0526_1 ↔ FGADR 1597_1 (1.00); FGADR train:0878_1 ↔ FGADR 0988_1 (1.00); FGADR val:0553_1 ↔ FGADR 0995_1 (1.00); TJDR_uwf train:TJDR_train_092 ↔ TJDR_uwf TJDR_test_021 (1.00); TJDR_uwf val:TJDR_train_101 ↔ TJDR_uwf TJDR_test_023 (1.00); Retinal-Lesions train:4428_left ↔ FGADR 0706_3 (0.99); Retinal-Lesions val:31746_right ↔ FGADR 0466_3 (0.98); Retinal-Lesions train:24235_right ↔ FGADR 0785_3 (0.98); Retinal-Lesions train:658_right ↔ FGADR 0236_3 (0.98); Retinal-Lesions train:15684_right ↔ FGADR 0627_3 (0.98). The full lists are bench/data/splits/leak_exclusions.json.

Supplementary Table 16 Released checkpoints on every test set, per class, under this protocol. Each checkpoint is run through its authors' own test pipeline (resize, normalisation, whole image or sliding window as configured) and the maps are scored with the protocol evaluator; a source the checkpoint was trained on is its internal test.

Checkpoint	Trained on	Test set	MA	HE	EX	SE	mAUPR
M2MRF	IDRiD	IDRiD	.408	.654	.832	.664	.639
M2MRF	IDRiD	DDR	.122	.289	.506	.061	.244
M2MRF	IDRiD	TJDR std	.294	.387	.566	.241	.372
M2MRF	IDRiD	TJDR UWF	.047	.066	.044	.096	.063
M2MRF	IDRiD	MAPLES	.220	.363	.457	.048	.272
M2MRF	IDRiD	FGADR	.010	.243	.210	.113	.144
M2MRF	IDRiD	Retinal-L.	.033	.136	.179	.068	.104
M2MRF	IDRiD	DiaRetDB1	.043	.247	.265	.107	.165
M2MRF	DDR	IDRiD	.276	.549	.678	.573	.519
M2MRF	DDR	DDR	.165	.426	.558	.456	.401
M2MRF	DDR	TJDR std	.300	.628	.664	.479	.518
M2MRF	DDR	TJDR UWF	.073	.512	.164	.075	.206
M2MRF	DDR	MAPLES	.204	.455	.429	.102	.297
M2MRF	DDR	FGADR	.015	.263	.312	.163	.188
M2MRF	DDR	Retinal-L.	.044	.148	.195	.188	.144
M2MRF	DDR	DiaRetDB1	.055	.234	.178	.247	.179
HRDecoder	IDRiD	IDRiD	.475	.699	.892	.763	.707
HRDecoder	IDRiD	DDR	.110	.355	.556	.148	.292
HRDecoder	IDRiD	TJDR std	.298	.578	.610	.426	.478
HRDecoder	IDRiD	TJDR UWF	.075	.413	.161	.359	.252
HRDecoder	IDRiD	MAPLES	.287	.444	.692	.083	.377
HRDecoder	IDRiD	FGADR	.016	.285	.338	.102	.186
HRDecoder	IDRiD	Retinal-L.	.038	.157	.338	.125	.165
HRDecoder	IDRiD	DiaRetDB1	.047	.329	.452	.175	.251
HRDecoder	DDR	IDRiD	.291	.639	.799	.633	.590
HRDecoder	DDR	DDR	.172	.557	.629	.486	.461
HRDecoder	DDR	TJDR std	.298	.707	.693	.517	.554
HRDecoder	DDR	TJDR UWF	.077	.543	.148	.102	.218
HRDecoder	DDR	MAPLES	.279	.471	.617	.159	.382
HRDecoder	DDR	FGADR	.013	.302	.291	.211	.204
HRDecoder	DDR	Retinal-L.	.064	.216	.283	.232	.199
HRDecoder	DDR	DiaRetDB1	.060	.409	.323	.306	.275

Supplementary Table 17 Foundation models without training, on every test set. Zero-shot automatic masks (density and area cap chosen on the validation pool), SAM 3 text prompts with the wording chosen on the validation pool, and prompts derived from the ground truth (an upper bound that reads the test labels). Values are mAUPR where the output carries classes and the any-lesion AUPR[a] where it does not.

Model	Setting	IDRiD	DDR	TJDR std	TJDR UWF	MAPLES	FGADR	Retinal-L.	DiaRetDB1
Zero-shot (any-lesion AUPR of the class-agnostic map; text prompts per class)									
SAM 2	zero_shot:auto_any_lesion	.023	.008	.012	.007	.003	.010	.014	.010
SAM 2	zero_shot:auto_oracle_class	.029	.009	.012	.007	.003	.010	.014	.010
SAM 2	zero_shot:auto_stability	.026	.011	.008	.018	.004	.011	.014	.008
SAM	zero_shot:auto_any_lesion	.020	.008	.008	.004	.003	.012	.014	.007
SAM	zero_shot:auto_oracle_class	.022	.008	.009	.005	.003	.013	.014	.007
SAM	zero_shot:auto_stability	.031	.014	.028	.015	.004	.020	.015	.009
Oracle boxes, every component									
MedSAM	oracle:box:all	.665	.648	.840	.860	.738	.827	.864	.714
SAM 2	oracle:box:all	.848	.756	.932	.922	.826	.825	.835	.771
SAM	oracle:box:all	.860	.772	.870	.869	.824	.759	.747	.777
Oracle points, one per component									
SAM 2	oracle:point:all	.116	.019	.081	.291	.009	.050	.058	.074
SAM 3, selected wording									
SAM 3	text:selected	.228	.125	.111	.185	.079	.108	.066	.088

[a] any-lesion AUPR of a class-agnostic map. Variant tokens as in the run records (density, cap, canvas).

Supplementary Table 18 Experiment names used in the text and the identifiers used in the repository. The run directories under runs/, the experiment column of data/results.csv and the planning documents in docs/ carry the identifiers; the text names each experiment by what it does.

Identifier	Name in the text	What it is	Where it is reported
L1	Paper audit	Extraction of the setup and the comparison tables of every method paper in the ledger	Fig. 2b,c; Supplementary Tables 1 and 2
L2	Reproducibility funnel	Registered fundus-specific methods by code, training code, weights and reproduced number	Fig. 2a
E1	Released checkpoints (author weights)	Released checkpoints scored with the paper's number, the authors' function and this protocol, on all test sets	Fig. 3a,b,d; Table 3a; Supplementary Table 16
E1r	Author-recipe reruns	The authors' own pipelines retrained from scratch on IDRiD (M2MRF, HRDecoder)	Fig. 3c; Table 3b
E2 (B-std)	Unified retraining; the unified protocol	Every tier-1 model with complete training code retrained on IDRiD and on DDR under the shared protocol	Fig. 4; Tables 4 and 6
E2c	Learning-rate sweep	HRNet-W48 and M2MRF on DDR with the recipe learning rate $\times 1/3$ and $\times 3$, judged on validation only	Extended Data Fig. 1
E3a	Independent tests	Every main-table checkpoint applied to the six test sets of sources never used for training	Fig. 4b–e; Table 6; Extended Data Fig. 3
E3b	Training-source experiment	Five representative models trained on the standard-field TJDR images and on FGADR	Fig. 6b–d; Supplementary Table 7
E4	Annotation-granularity experiment	U-Net and M2MRF on fine, mixed and coarse training labels, alone and combined	Fig. 6e; Table 7; Supplementary Table 7
E5	Foundation models under four kinds of use	Zero-shot, oracle prompts, parameter-efficient fine-tuning, encoder + decoder; the encoder comparison	Fig. 5; Table 5; Supplementary Tables 10 and 17
E7	Input-resolution experiment	M2MRF and RETFound + UPerNet at a 512- or 1,440-pixel canvas and on native-resolution patches	Fig. 7d; Table 7
E8	Training-amount experiment	U-Net and M2MRF on 10%, 50% and 100% of DDR and on the merged fine-grained sources	Fig. 7e; Table 7
E9	Loss experiment	U-Net on DDR with four losses, architecture fixed	Extended Data Fig. 2; Table 7
E10	Computational cost	Parameters, trainable parameters and GPU hours of every run; the inference audit	Extended Data Fig. 4; Supplementary Table 8
ENS	Ensembles	Pixelwise means of the probability maps in the registered compositions	Table 6; Supplementary Table 11
REF	Label revision (Refined IDRiD)	IDRiD-trained checkpoints re-scored against the refined and the official labels	Supplementary Table 9
HUMAN	Human reference	Each DiaRetDB1 expert against the consensus of the other three	Table 6; Supplementary Table 13
CLIN	Results by DR grade	Every score recomputed within grades 0–1 and 2–4	Supplementary Table 12
N8	Reference model (RetiSeg-Ref)	Two models trained on the merged fine-grained sources and averaged	Methods; Supplementary Table 11

Supplementary Data

The following files will be released with the code.

- **Supplementary Data 1.** Split lists of every source (`bench/data/splits/` in the code repository): the image identifiers of each train, validation and test split, the subset lists of the training-amount experiment, and the near-duplicate exclusions.
- **Supplementary Data 2.** The literature extraction (`literature/meta/reported_results.tsv`): one row per paper, baseline, dataset, class and metric, with the stated provenance of every number, and the per-paper audit (`paper_audit.tsv`).
- **Supplementary Data 3.** The results table behind every figure and table of the main text and of this document (`docs/paper/retiseg_paper/paper/data/results.csv`), with the recipe, environment and deviation record of every run; its experiment column uses the identifiers of Supplementary Table 18.
- **Supplementary Data 4.** Probability maps of every main-table model on every test set whose licence allows derived data to be released (IDRiD, DDR, TJDR, MAPLES-DR, DiaRetDB1), as 16-bit images, so that any metric can be recomputed.